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CHM 428 — Past Questions & Materials
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Institution: Federal University of Petroleum Resources, Effurun
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Paper 1 of 4 — CHM 428 - Selected Topics in Industrial Chemistry

Level 400 • First Semester • PastQuestion

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Federal University of Petroleum Resources, Effurun
Department of Chemistry
B.Sc (Hons.) Degree Examination, First Semester 2024/2025 Session; Sept. 2025
CHM 428- Selected Topics In Industrial Chemistry (2 credits)

Answer any Three questions only. Time Allowed — 2 hours

Question 1

- a) Sodium-ion batteries are considered a potential alternative to lithium-ion batteries. Discuss. [9 marks]
- b) Define a battery and explain the difference between a primary and secondary battery. Give two examples of each type. [6 marks]
- c) Explain three key performance parameters of batteries. [3 marks]
- d) List the components of lithium-ion battery with their functions. [6 marks]

Question 2

- a) Explain the Bingel reaction. Why is it significant in fullerene chemistry? [10 marks]
- b) Describe the structure of Buckminsterfullerene (C_{60}). [5 marks]
- c) Explain how its structure contributes to stability and aromaticity. [3 marks]
- d) Outline two major methods for large-scale production of fullerenes. State one advantage of each. [6 marks]

Question 3

- a) Differentiate between commodity chemicals, specialty chemicals, and fine chemicals, giving two examples each. [12 marks]
- b) Explain two key characteristics that distinguish fine chemicals from specialty chemicals. [6 marks]
- c) Discuss the role of these three categories of chemicals in the value chain of the chemical industry. [6 marks]

Question 4

- a) Explain the concept of retrosynthesis, using ibuprofen as an example. [8 marks]
- b) Outline the importance of stereochemistry and enantioselective synthesis in fine chemical production, with one example. [8 marks]
- c) Compare homogeneous, heterogeneous, and biocatalysis in fine chemical synthesis, giving one advantage of each. [8 marks]

END OF PAPER

Paper 2 of 4 — CHM 428 PQ Selected Topics in Industrial Chemistry 2023

Level 400 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN, DELTA STATE
 COLLEGE OF SCIENCE
 DEPARTMENT OF CHEMISTRY
 SECOND SEMESTER EXAMINATION 2022/2023 SESSION SEPTEMBER, 2023
 COURSE TITLE: SELECTED TOPICS IN INDUSTRIAL CHEMISTRY
 COURSE CODE: CHM428
 INSTRUCTION: ANSWER FOUR QUESTIONS ONLY TIME: 3 HOURS

QUESTION ONE

1. (a) Explain the main principles governing environmental forensic investigation. 9marks
- (b) Discuss four key parameters that can be used in monitoring of pollution during environmental forensic investigation. 9marks
- b) Explain four analytical instrument that can be used in environmental forensic studies
- (c) Outline the steps that can be taken for an environmental forensic assessment of crude oil pollution in Ogoni land/Community. 7marks OR
- (d) Explain the role of an environmental forensic expert in the assessment of pollution in any community where pollution is suspected to have occurred. 7marks

25MARKS

QUESTION TWO

- (a) Nigeria holds the seventh largest natural gas reserve of over 200 trillion cubic feet in the world. How will the reserve be harnessed to realize the sustainable development goal number one 10marks
- (b) Explain three key factors that may impair the realization of this goal. 6marks
- (c) Briefly explain the Nigeria's gas master plan. 9marks

25MARKS

QUESTION THREE

- (a) Briefly explain the chemistry of fullerenes with reference to the following:
 - (i) Structure (ii) Types (iii) Addition of carbon nucleophiles to fullerenes
 - (iv) Pharmaceutical applications. 10marks
- (b)i Aluminium-air battery is an attractive candidate for metal-air batteries. Discuss the difference between primary batteries and secondary batteries? 8marks
- ii. What are the four major parts of lithium batteries? 8marks
- (d) Discuss the applications of metal organic frameworks (MOFs) in lithium ion batteries 7marks

25MARKS

Paper 3 of 4 — CHM 428 Handout 1 and 2

Level 400 • Second Semester • Handout

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Fullerenes: Chemistry and its Applications

Madhura Mohan Gokhale¹ and Rakesh Ravindra Somani^{2,*}

¹Department of Pharmaceutical Chemistry, Vivekanand Education Society's College of Pharmacy, Chembur, Mumbai, India; ²Professor & Head of Department of Pharmaceutical Chemistry, Vivekanand Education Society's College of Pharmacy, Chembur, Mumbai, India

Abstract: Fullerenes being allotropes of carbon, have been considered as new class of molecules. Unlike diamond and graphite, this is made up of hollow carbon cage structure. The idea of spherical cage structures of C_{60} arose from construction of geodesic domes made by renowned architect Buckminster Fuller. Although fullerenes have low solubility in physiological media they find promising biological applications. The photo, electrochemical and physical properties of C_{60} and other fullerene derivatives finds applications in medical fields. The ability of fullerenes to fit inside the hydrophobic cavity of HIV proteases makes them potential inhibitor for substrates to catalytic active site of enzyme. It possesses radical scavenging and antioxidant property. At the same time, when it exposed to light it can form singlet oxygen in high quantum yields which with direct electron transfer from excited state of fullerenes and DNA bases finally results in cleavage of DNA. The fullerenes are also used as a carrier for gene and drug delivery system. The associated low toxicity of fullerenes is sufficient to attract the researchers for investigation of these interesting molecules.

Keywords: Applications, buckyball, C_{60} , drug delivery, fullerene, toxicity.

1. INTRODUCTION

Carbon, the common element and widely distributed in nature, is known to exist in several forms viz. graphite and diamond. Fullerenes are fourth allotropic form of carbon. In comparison with graphite and diamond with extended solid state structures, fullerenes are spherical molecules which have solubility in various organic solvents. This property can be used for different chemical manipulation.

A fullerene is a carbon cage structure having fused ring system which consists of pentagons and hexagons. The first proposal of buckyball was given by Eiji Osawa, Japan. He recognized that corannulene, a cyclopentane ring fused with 5 benzene rings was a part of football framework and hypothesized that entire structure could exist. In 1985, the group of scientists Richard Smalley, Robert Curl, James Heath, Sean O'Brien, and Harold Kroto [1] synthesized the first fullerene molecule, buckminsterfullerene (C_{60}) at Rice University. These scientists named newly discovered molecule in honor of architect R. Buckminster Fuller who created geodesic dome with same shape.

Fullerenes are composed of fused hexagons and pentagons (Fig. 1). C_{60} and C_{70} are most accessible members of this family. The high symmetry found in this molecule is an important property. C_{60} molecule is composed of 60 carbon atoms which are arranged as 12 pentagons and 20 hexagons [2]. All rings are fused and double bonds are highly conjugated. The reason for high symmetry of this molecule is 120 symmetrical operations found in molecule like rotation around the axis and reflection in plane which

maps the molecule onto itself. When a carbon atom is placed at each vertex of molecule which all valences satisfied by two single bonds and one double bond, stabilized by resonance gives structure of C_{60} molecule, appears to be aromatic.

Other forms [3] of fullerene are Fullerane which is fully saturated fullerene (example, the hydrocarbon $C_{60}H_{12}$) and Fullerooids are fullerene-like compounds as they resemble fullerenes in structure but do not confirm the definition of a fullerene (example, Heterofullerenes, Norfullerenes, Homofullerene, Secofullerene).

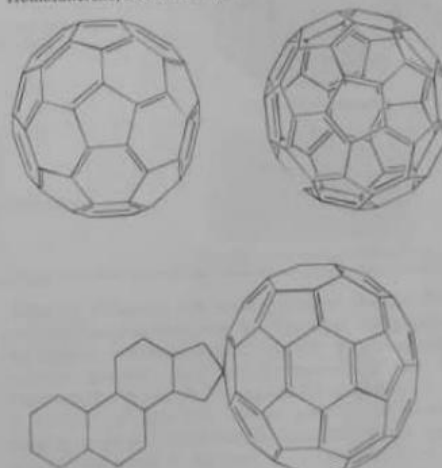
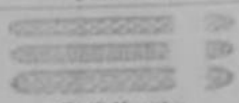
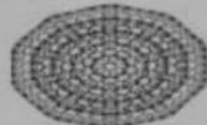
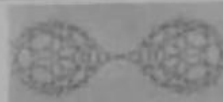


Fig. (1). Different fullerene derivatives.

*Address correspondence to this author at the Department of Pharmaceutical Chemistry, Vivekanand Education Society's College of Pharmacy, Chembur, Mumbai, India; Tel: +91 9833771384; E-mail: rakeshsomani@gmail.com

2. TYPES OF FULLERENES

Table 1. Types of fullerenes.

No. no	Types	Description	Representation
1.	Buckyball clusters [4] (Fig. 2)	Smallest number is C_{36} and most common is C_{60} . Because sphere of buckyball is hollow, other atoms can trapped within it.	 Fig. (2). Buckyball clusters
2.	Nanotubes [5] (Fig. 3)	Hollow tubes of very small dimensions with single or multiple walls, used in electronics industry.	 Fig. (3). Nanotubes
3.	Mega tubes [6] (Fig. 4)	Varying in dimensions than nanotubes as larger in diameter with walls of different thickness; application in transport of a variety of molecules of different sizes.	 Fig. (4). Mega tubes
4.	Polymers [7] (Fig. 5)	Chain, two or three dimensional, formed under high pressure and temperature conditions.	 Fig. (5). Polymers
5.	Nano 'onions' [8] (Fig. 6)	Spherical particles having multiple layers which surrounds buckyball core. Typical diameter is 3-5 nm; proposed for lubricants.	 Fig. (6). Nano 'onions'
6.	Linked 'ball-and-chain' dimers [9] (Fig. 7)	Two buckyballs linked by a carbon chain.	 Fig. (7). Linked ball and chain dimers.

3. GENERAL PROPERTIES OF FULLERENES

Pure fullerenes have better close packing than impure one. Derivatives of C_{60} show range of solubility [10]. Fluorinated derivatives are more soluble and some bromo derivatives are much less soluble. Fullerenes have low density (1.65 g/cc) relative to diamond (3.51 g/cc). Fullerenes are stable upto temperature of 1000°C. As the cage is entirely of sp^2 -hybridized carbons it gives electron withdrawing negative inductive (-I) effect so fullerenes are strongly electron-attracting i.e. (that is) fullerenes can readily react with nucleophiles. High strain energies contribute to

reactivity. Fullerenes can undergo various reactions [11] such as reduction, oxidation, hydrogenation, halogenation, nucleophilic reactions, radical reaction, transition metal complex reaction, regioselective reaction and so on. Fullerenes are susceptible to degradation or decomposition in presence of light [12] and oxygen. The intersystem crossing of singlet excited state to energetically lower triplet excited state results in decaying of C_{60} . The triplet excited states are prone to many deactivation processes such as ground state quenching, quenching by molecular oxygen and transfer of electrons to molecules.

4. STABILITY OF FULLERENES

Localization of electrons and ring strain affects stability of fullerenes. By applying huckel rule [13] to different fullerene derivatives like C_{50} , C_{60} , C_{70} , C_{82} and C_{90} , among these only C_{50} and C_{70} were predicted to be aromatic and C_{60} was less aromatic. As Huckel's rule gives aromaticity in spherical systems than planar systems, only C_{60} was found to be more aromatic and highly stable. The highest stability of C_{60} is due to reduced strain in ring as twelve pentagons are isolated from each other [14]. When there are minimum bond orders in five-membered rings it indicates that all double bonds are in six-membered rings and none in five-membered rings. When meta position (i) of any two rings are fused with hexagonal ring, as in Fig. (8), this arrangement permits minimization of bond orders in five-membered rings. The degree of instability in C_{60} will occur when requirement of minimization of bond orders is not satisfied as there is para relationship between two pentagonal rings. It is significantly seen in higher fullerenes like C_{70} , it is less stable than C_{60} which contains five arrangements of type (ii). This indicates that as number of carbon atoms increases, stability is going to decrease.

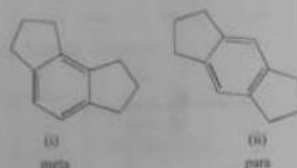


Fig. (8). Stability of fullerenes.

5. FULLERENE PRODUCTION

There are various methods for fullerene production such as, 1) Electric Arc method, 2) Arc heating method, 3) Laser ablation method and 4) Formation of Fullerenes by pyrolysis of Hydrocarbons. These methods are generally used for large scale production.

5.1. Electric Arc Method

Equipment bench top reactor by Wudl [15] (Fig. 9) was used for production of fullerenes in this method. The

advantage is simple construction of reactor, low cost which makes them attractive for synthetic chemists [16]. A current (AC or DC) of 40-60 A is applied to thin graphite rod supported by copper sleeve having sharpened tip results in evaporation of material of thin rod. The rod is consumed till the point where rod cannot make contact with the thick rod. Then power is shut off by this technique, 5-10% of fullerene is obtained.

(A) Vacuum system, (B) Pyrex jelly bar, (C) graphite rod(1 mm), (D) graphite rod(4 mm), (E) copper electrode, (F) manometer

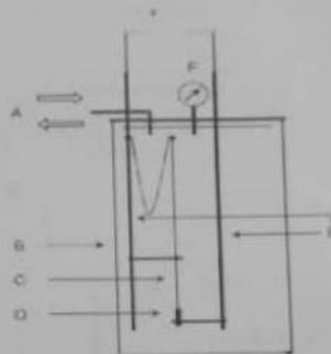


Fig. (9). Apparatus used by Wudl.

5.2. Arc Heating of Graphite

This technique is developed by Smalley [17] (Fig. 10) which is used as an alternative to arc vaporization of graphite. There is a dissipation of electrical power in arc when the tips of two sharpened graphite rods are kept in close proximity but they are not in direct contact with other. When the electrodes are barely touching, known as "contact-arcing" [18] this technique works more efficiently. This method also allows an efficient evaporation of carbon. The yield of obtained fullerenes by this process was found to be about 15%.

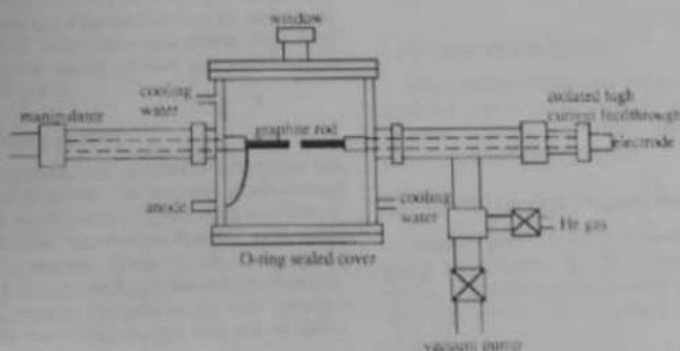


Fig. (10). Arc heating of graphite.

4 Mini-Reviews in Organic Chemistry, 2015, Vol. 12, No. 4

Gebriale and Semani

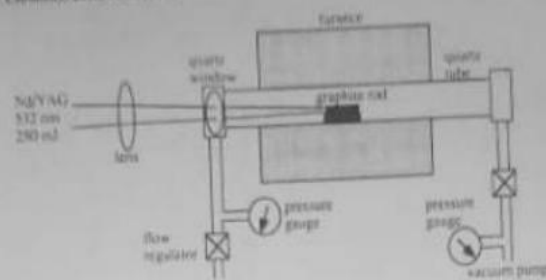


Fig. (11). Laser ablation.

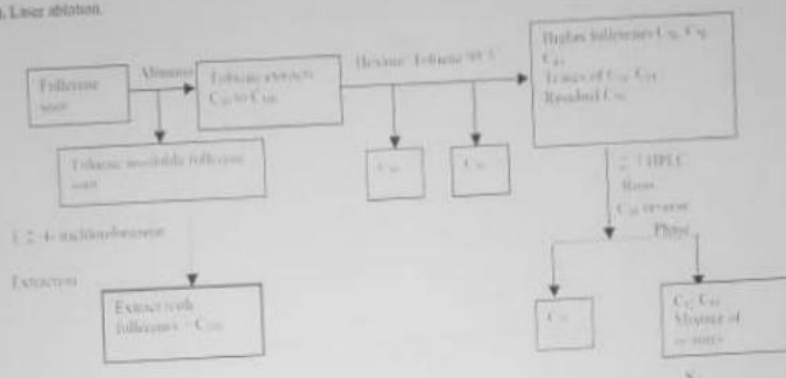


Fig. (12). Scheme for separation and purification.

5.3. Laser Ablation [19, 20]

It is very powerful and useful technique for production of fullerene clusters. It involves laser ablation of graphite in helium atmosphere. Laser ablation is the process which involves removal of material from a solid (or occasionally liquid) surface with the help of irradiation using laser beam. As in this technique solvents are not used and operators are not exposed to chemicals, it is considered as environmentally friendly process. The material used in this process generally gets converted to plasma at higher laser flux. By carrying out this process at high temperature, this plasma cools more slowly and generated carbon clusters gets rearranged in stable fullerenes (Fig. 11).

5.4. Pyrolysis of Hydrocarbons

Pyrolysis of hydrocarbons (preferably aromatics) can also be used to get fullerenes. The pyrolysis of naphthalene at 1000° C in an argon stream [21] was done first time to obtain fullerenes. The naphthalene skeleton is a small fragment of the C_{60} structure. The bowl-shaped corannulene [22] was used as precursors for C_{60} , followed by naphthalene. Synthesis of fullerenes is also possible by laser pyrolysis which involves the use of benzene and acetylene as carbon source.

6. SEPARATION AND PURIFICATION OF FULLERENE [23]

The first step involves extraction of soluble fullerenes from the crude carbon soot. Benzene was first used for extraction, but for safety reasons, it is replaced with toluene. Secondly, it is isolated by chromatographic separation using alumina as stationary phase and either pure hexane or hexane-toluene (95:5) as mobile phase. The general scheme is shown in Fig. (12).

7. CHARACTERIZATION

The separation techniques can provide pure C_{60} or C_{70} molecules, but it is also important to verify the purity of obtained compounds. The techniques used for characterization of fullerenes are

7.1 ^{13}C NMR (Nuclear Magnetic Resonance) [24, 25]

Solution phase ^{13}C NMR is powerful probe of fullerene purity and structure. All 60 carbon nuclei are equivalent in C_{60} . ^{13}C NMR spectrum exhibit only one single peak in Fig. (13) whereas in C_{70} exhibit five distinct resonance (Fig. 13).

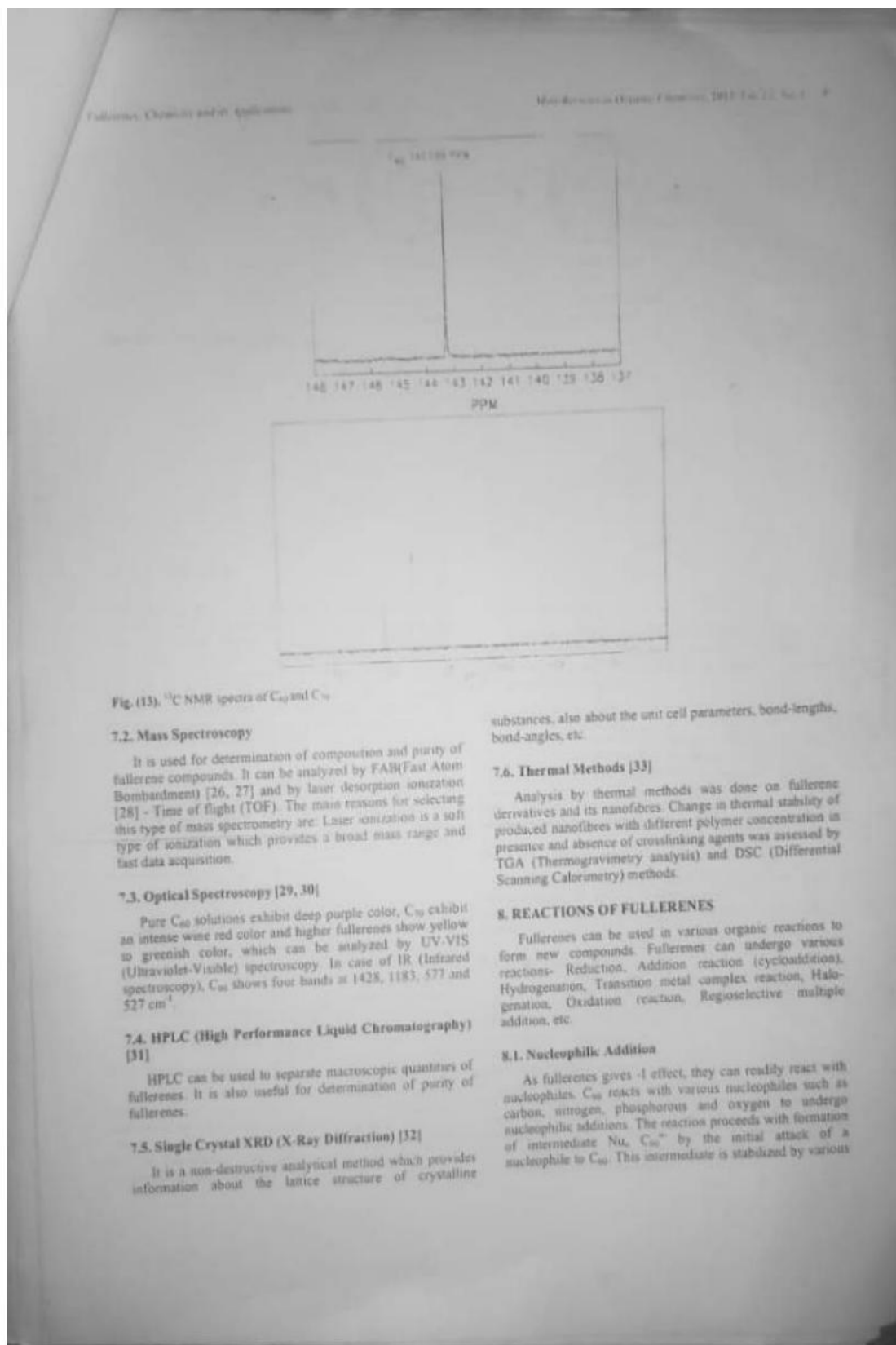


Fig. (13). ^{13}C NMR spectra of C_{60} and C_{70}

7.2. Mass Spectrometry

It is used for determination of composition and purity of fullerene compounds. It can be analyzed by FAB (Fast Atom Bombardment) [26, 27] and by laser desorption ionization [28] - Time of flight (TOF). The main reasons for selecting this type of mass spectrometry are: Laser ionization is a soft type of ionization which provides a broad mass range and fast data acquisition.

7.3. Optical Spectroscopy [29, 30]

Pure C_{60} solutions exhibit deep purple color, C_{70} exhibit an intense wine red color and higher fullerenes show yellow to greenish color, which can be analyzed by UV-VIS (Ultraviolet-Visible) spectroscopy. In case of IR (Infrared spectroscopy), C_{60} shows four bands at 1428, 1183, 577 and 527 cm^{-1} .

7.4. HPLC (High Performance Liquid Chromatography) [31]

HPLC can be used to separate macroscopic quantities of fullerenes. It is also useful for determination of purity of fullerenes.

7.5. Single Crystal XRD (X-Ray Diffraction) [32]

It is a non-destructive analytical method which provides information about the lattice structure of crystalline

substances, also about the unit cell parameters, bond-lengths, bond-angles, etc.

7.6. Thermal Methods [33]

Analysis by thermal methods was done on fullerene derivatives and its nanofibres. Change in thermal stability of produced nanofibres with different polymer concentration in presence and absence of crosslinking agents was assessed by TGA (Thermogravimetry analysis) and DSC (Differential Scanning Calorimetry) methods.

8. REACTIONS OF FULLERENES

Fullerenes can be used in various organic reactions to form new compounds. Fullerenes can undergo various reactions- Reduction, Addition reaction (cycloaddition), Hydrogenation, Transition metal complex reaction, Halogenation, Oxidation reaction, Regioselective multiple addition, etc.

8.1. Nucleophilic Addition

As fullerenes gives -I effect, they can readily react with nucleophiles. C_{60} reacts with various nucleophiles such as carbon, nitrogen, phosphorous and oxygen to undergo nucleophilic additions. The reaction proceeds with formation of intermediate $\text{Nu}_n\text{C}_{60}^-$ by the initial attack of a nucleophile to C_{60} . This intermediate is stabilized by various

6 Mini-Reviews in Organic Chemistry, 2015, Vol. 12, No. 4

Gokhale and Sonawale

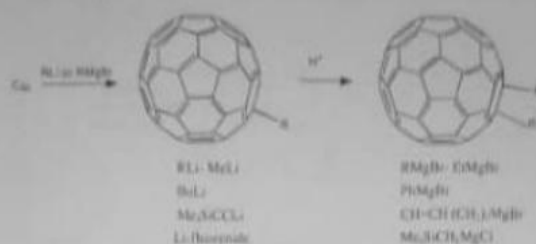
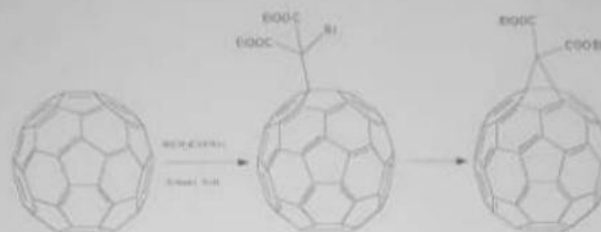
Fig. (14). Addition of nucleophile on C_{60} 

Fig. (15). Bingel reaction

reactions such as (a) the addition of electrophiles E , (b) addition of neutral electrophiles EX such as alkylhalogenides and cyclohexenofullerenes, or by (d) an oxidation. Nucleophilic addition reactions are of various types, such as addition of carbon nucleophiles, addition of cyanide, addition of hydroxides and alkoxides, addition of phosphorus, silicon, germanium nucleophiles and addition of macromolecular nucleophiles as in case of fullerene polymers.

8.1.1. Addition of Carbon Nucleophiles

1) C_{60} can react with Organolithium ($R-Li$) and Grignard compounds ($R-Mg-X$) using alkyl, phenyl or alkaryl compounds to form primary intermediates as the anions RC_{60}^- . To obtain maximum yield, organolithium compounds as nucleophiles are used [34]. Different Grignard reagents can be used, given in Fig. (14).

2) Bingel reaction [35]: It is a classic example of nucleophilic addition reaction which was first discovered by Bingel, 1993 using bromo derivative of diethyl malonate in presence of a sodium hydride as base to form methanofullerenes (Fig. 15). The reaction mechanism involves abstraction of proton by base to form carbanion or enolate which reacts with electron deficient fullerene double bond as shown in Fig. (15), followed by displacement of bromine in an intramolecular ring closure.

8.1.2. Addition of Cyanide [36]

The negative shift of reduction potential results due to addition of alkyl nucleophile. To compensate this potential addition of strong electron withdrawing group is necessary such as addition of cyanide. C_{60} reacts with $LiCN$ or $NaCN$ to form monoadduct anion which can be quenched with various nucleophiles (Fig. 16).

8.2. Cycloaddition

As double bonds of C_{60} exhibit a dienophilic character various cycloaddition reactions are possible. The reactivity of the diene is mainly responsible for the various conditions for cycloadduct formation.

8.2.1. [4+2] Cycloaddition

Many [4+2] cycloaddition reactions with C_{60} are done under thermal conditions. The reaction of cyclopentadiene [37] and C_{60} gives the monoadduct in comparatively high yield at room temperature as shown in Fig. (17). The anthracene cycloadduct is formed when excess of diene is refluxed in toluene (Fig. 17).

8.2.2. [3+2] cycloaddition

To obtain pyrazolone intermediate C_{60} reacts with diazomethane in toluene. Extrusion of N_2 from this intermediate results in two different bridged fullerenes [38], as shown in Fig. (18).

8.3. Hydrogenation [39]

Hydrogenation of fullerenes can be done easily and by various processes. $C_{60}H_{14}$ and $C_{60}H_{16}$ are common examples of hydro fullerenes. Due to large strain in ring completely hydrogenated fullerene, $C_{60}H_{60}$ is hypothetical. The prolonged hydrogenation by direct reaction with hydrogen gas at high temperature results in cage formation which leads to instability of highly hydrogenated fullerenes. Fullerenes are easily hydrogenated by several methods. Examples of hydro fullerenes are $C_{60}H_{14}$ and $C_{60}H_{16}$. Completely hydrogenated $C_{60}H_{60}$ is only hypothetical because of large strain. Finally there is collapse of cage structure which forms polycyclic aromatic hydrocarbons.

Fullerenes: Chemistry and its Applications

Mini-Reviews in Organic Chemistry, 2015, Vol. 12, No. 4 5

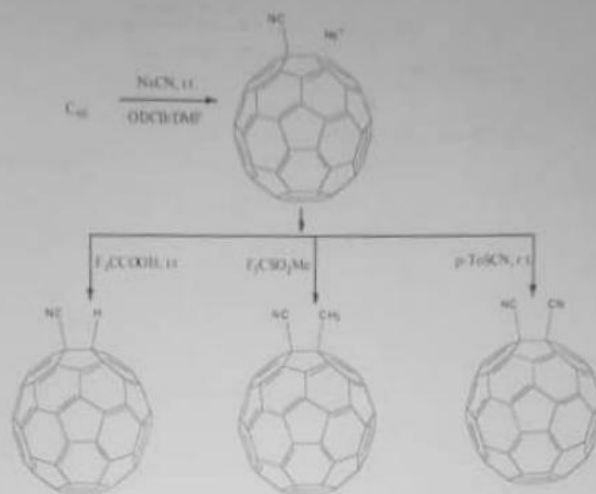


Fig. (16). Addition of cyanide.

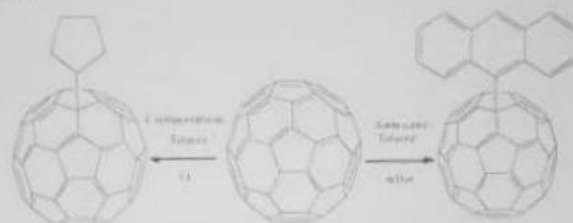


Fig. (17). [4+2] cycloaddition

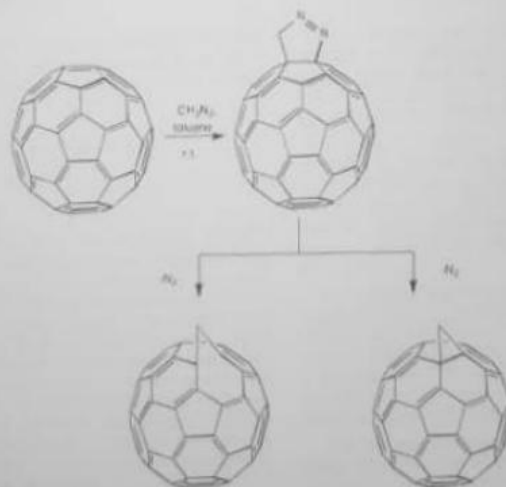


Fig. (18). [3+2] cycloaddition

Mini-Reviews in Organic Chemistry, 2015, Vol. 12, No. 4

Gokhale and Jomani

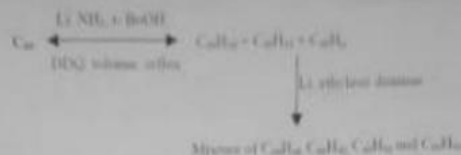


Fig. (19). Birch-Huckel reduction.

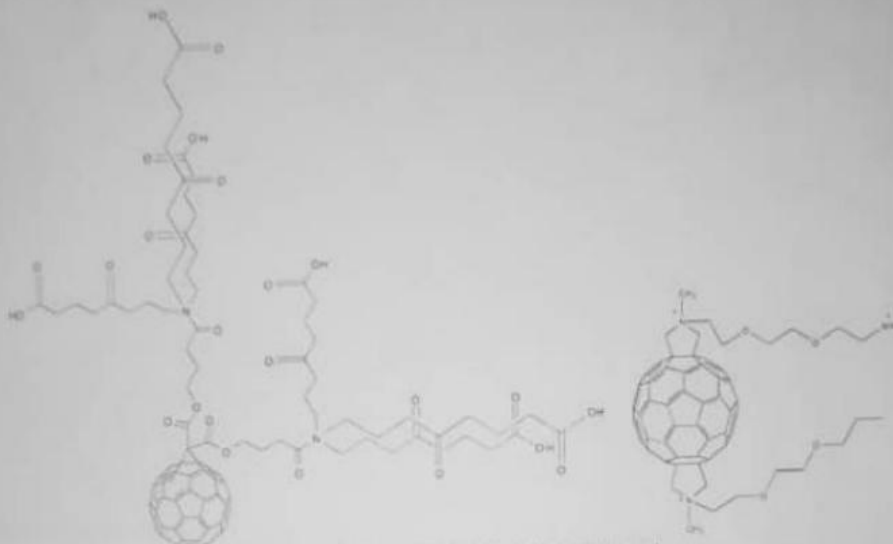


Fig. (20). Fullerenes derivatives

E.g. (Example) - Birch-Huckel Reduction, Fig. (19) Hydrogenation of C_{60} was done with Lithium (Li) in liquid ammonia (Liq NH_3). The major products of this are the isomers of $\text{C}_{60}\text{H}_{36}$ and $\text{C}_{60}\text{H}_{24}$ (Fig. 19)

9. APPLICATIONS OF FULLERENES

9.1. Biological Applications

9.1.1. Antiviral Activity

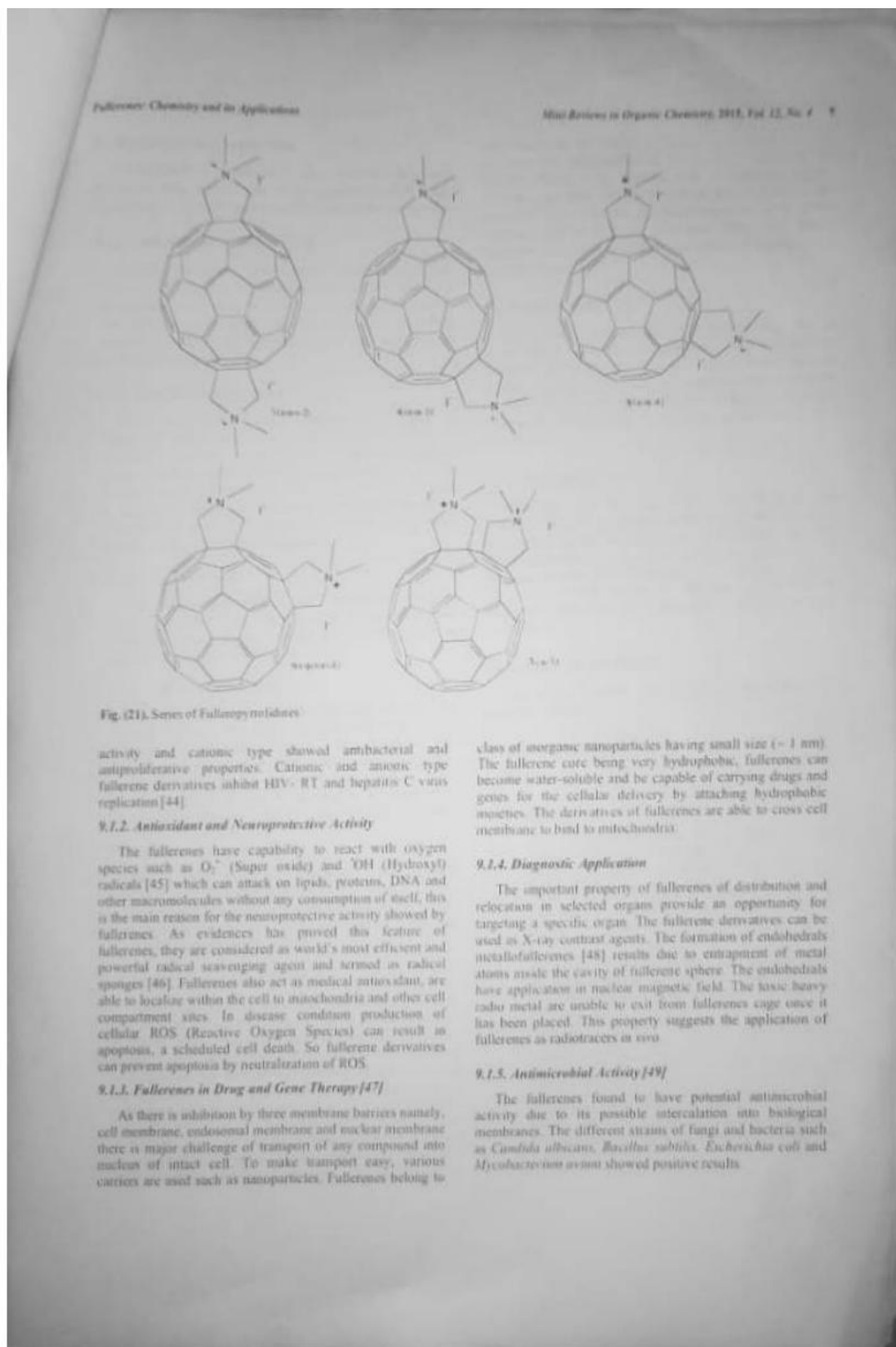
Although tremendous advances in the therapy of AIDS are done, there are several major problems associated with current antiviral drugs and its treatment such as, quick mutation of Human HIV (Human Immunodeficiency Virus), developed resistance and high toxicity leads to urgent need to develop new antiviral drugs. HIVP (HIV Protease) is basic and important enzyme required for the virus survival which is an aspartic protease enzyme, specific for HIV proteins and does not react directly or indirectly with human proteases. The activation of RT (Reverse Transcriptase), RNase, Integrase and Protease is done by the cleavage of polyprotein by HIVP. The HIVP active site is a cylindrical hydrophobic cavity having diameter of around $10 \text{ \AA}$ which has two amino acidic residues, 25 and 125 aspartate [40].

It was suggested that fullerene derivatives acts as anti-HIV agents by fitting smartly in the active site of the viral protease due to strong Van der Waals interactions on hollow surface of enzyme. The hydrophobic interactions between C_{60} and regions of cavity plays role in holding the inhibitor tightly [41].

Earlier studies have showed that derivatives of fullerenes can inhibit HIVP and can form complex with the same (Fig. 20). As shown in Fig. (20), Dendrofullerene 1 was found to have the highest anti-protease activity and Derivative 2 which is trans-2 isomer strongly inhibits the replication of HIV-1 [42].

Fulleropyrrolidines having two ammonium groups was found to be active against HIV-1 and HIV-2. A series of fullerene derivatives (Derivatives 3-7, Fig. 21) was synthesized to elucidate the structural parameters which affect the antiviral activity. The results showed that trans fullerene derivatives are more active than cis where the equatorial fullerene derivative is totally inactive [43] (Fig. 21).

The fullerene derivatives of cationic and anionic type inhibit the replication of HIV-RT and hepatitis C virus. Anionic type of fullerenes was found to possess antioxidant



9.2. Pharmaceutical Application

CNTs (Carbon Nanotubes) - other allotropes of carbon having structure in nano scale with length-to-diameter ratio greater than 1, 00, 000 known as CNTs (Carbon Nanotubes).

9.2.1. Carrier for Drug Delivery [50]

Carbon nanotubes with different functionalities are reported for targeting of Amphotericin B to Cells. The intracellular penetration was enhanced when doxorubicin with nanotubes have given. Due to their nanosize carbon nanotubes finds an application in tablet manufacturing as lubricants or glidants.

9.2.2. Genetic Engineering [51]

CNT and CNH (Carbon nanohorns) are used to manipulate genes and atoms in the development of biomaging genomes, proteomics and tissue engineering in genetic engineering. Nanotubes and nanohorns have ability to adhere various antigens on their surface which can be used as source of antigen in vaccines. Therefore the use of nanotubes as source of antigen can avoid use of dead bacteria which is sometimes dangerous.

9.2.3. Preservative [52]

Their antioxidant property can be used to preserve formulations which are prone to oxidation. They are used as preservative in anti-aging cosmetics and with zinc oxide in formulations to prevent oxidation.

9.2.4. Artificial Implants [53]

Generally the body exhibits rejection reaction for implants with the post administration pain. This can be overcome by use of immature sized nanotubes and nanohorns get attached with other proteins and amino acids. Due to their high tensile strength, carbon nanotubes filled with calcium and arranged or grouped in the structure of bone can act as bone substitute or in form of artificial joints.

9.2.5. Diagnostic Tool [54, 55]

Due to their ability of fluorescence with specific biomolecules, protein-encapsulated or protein filled nanotubes have been tried as implantable biosensors. For the study of cells and biological systems. Nanosize robots and motors with nanotubes can be used.

10. TOXICITY OF FULLERENES

There are number of factors that may be responsible for toxicity of fullerenes including its crystal structure, surface modifications and preparation methods. Generally greater the solubility of fullerene sample, lesser the toxicity associated with exposure. But these above mentioned factors may drive the water solubility of fullerenes [56]. As per the review of Fathi mousa [57] shows that pristine C_{60} is not relatively toxic. The evidences have been showed that fullerenes have no acute or sub-acute toxicity in large population of living organisms. Various risk assessment programs [58] have been done to evaluate toxicity of fullerenes. The conclusions of

such programs are based on limited amount of identified exposure and uncertainties of the quality of exposure and toxicity. There is no suitable data obtained for chronic exposure and their endpoints such as carcinogenicity, etc. Therefore, it is unrealistic to make any generalizations about exposure and toxicity of fullerenes. Also, new testing parameters need to be invented for conscious evaluation of toxicity of fullerenes.

11. FUTURE ASPECTS

The unique applications of fullerenes are due to extraordinary structure of fullerenes. Due to its poor solubility in common organic solvents, spectroscopic analysis of products, purification, separation and assessment of purity will become quite difficult. The high cost and low availability of fullerenes necessitates working on a very small scale.

12. CONCLUSION

Over the past few years, many new findings and important aspects of these carbon molecules have been accumulated to develop a new exciting scientific field. The direct application of fullerene and their derivatives to biological targets finds promising applications in medicine. The associated low toxicity of fullerenes is sufficient to attract the researchers for investigation of these interesting molecules.

CONFLICT OF INTEREST

The authors confirm that this article content has no conflict of interest.

ACKNOWLEDGEMENTS

Declared none

REFERENCES

- [1] Kroto, H.W.; Heath, J.R.; O'Brien, S.C.; Curl, R.F.; Smalley, R.E. C₆₀ Buckminsterfullerene. *Nature*, **1985**, *318*, 352.
- [2] Taylor, R.; Walton, R.M. The chemistry of fullerenes. *Nature*, **1993**, *363*, 685.
- [3] Powell, W.H.; Cross, F.; Moss, G.P.; Hodgson, C.; Hwa, R.J.R.; Yarn, A. Nomenclature for the C₆₀, B and C₇₀ (D5h) Fullerenes (IUPAC Recommendations). *Pure Appl. Chem.*, **2002**, *74*, 832.
- [4] Moxley, G.L.; Tate, D.A. *Inorganic Chemistry*, 3rd Ed. Pearson Education, **2004**.
- [5] Wang, X.; Li, Q.; Xu, J.; Jin, Z.; Wang, J.; Li, Y.; Jiang, K.; Fan, S. Fabrication of Ultralong and Electrically Uniform Single-Walled Carbon Nanotubes in Chem Substrates. *Nano Lett.*, **2009**, *9*(9), 3157-3161.
- [6] Michel, D.R.; Brown, R.M.; Spiess, T.L.; Romanova, D.K.; Lagew, R.J. The Synthesis of Megatubes: New Dimensions in Carbon Materials. *Inorg. Chem.*, **2001**, *40*(12), 2751-2755.
- [7] Hsiao, B.C.; Chou, T.; Borbone, E.; Khoshdel, A.; Baybordi, H.; Vignos, L.; Crespi, H. Synthesis of Donor-Acceptor Multiblock Copolymers Incorporating Fullerene Backbone Repeat Units. *Macromolecules*, **2010**, *43*(14), 6033-6044.
- [8] Saito, N.; Wang, H.; Chidswalla, M.; Alexandru, I.; Aronchberg, G.A.J. Synthesis of carbon 'nanowires' in water. *Nature*, **2001**, *414*(6863), 506-507.

Paper 4 of 4 — CHM 428: Specialty and Fine Chemicals Production

Level 400 • Second Semester • Handout

CHM 428: Specialty and Fine Chemicals Production

Course Description:

This course provides an in-depth study of the production, processing, and application of commodity, specialty and fine chemicals, focusing on their industrial synthesis, quality control, environmental considerations, and market significance. Emphasis is placed on low-volume, high-value chemical products used in pharmaceuticals, agrochemicals, dyes, cosmetics, electronics, and related industries.

Course Objectives:

By the end of the course, students will be able to:

- Distinguish between commodity, specialty, and fine chemicals.
- Understand synthetic methods used in fine chemical production.
- Describe the synthesis, purification, and quality assurance of specialty and fine chemicals.
- Analyze production techniques, raw materials, and process optimization as well as economic and regulatory factors, and sustainability issues.
- Discuss quality, safety, regulatory, and environmental concerns in specialty chemicals manufacture.
- Explore local and global industrial trends and opportunities.
- Apply knowledge to real-world case studies in pharmaceuticals, agrochemicals, materials, and consumer products.
- Develop skills in R&D, custom manufacturing, and quality control relevant to the chemical industry.

Course Modules and Topics

Module 1: Introduction to Specialty and Fine Chemicals

- Definitions and distinctions
- Characteristics of specialty and fine chemicals vs. commodity chemicals
- Industrial and economic significance
- Market trends and global players
- Nigerian context and development potential

Module 2: Classification and Functional Groups of Specialty Chemicals

- Pharmaceuticals and drug intermediates
- Agrochemicals: pesticides, herbicides, insecticides
- Flavours, fragrances, and food additives
- Electronic chemicals and photoresists

- Cosmetic and personal care chemicals
- Industrial water treatment and functional polymers
- Textiles, Dyes, pigments, and optical brighteners

Module 3: Fine Chemicals Production

- Fundamentals of Fine Chemical Synthesis Production of Active Pharmaceutical Ingredients (APIs)
- Other Fine Chemical Categories

Module 4: Specialty Chemicals Production and Applications

- Formulation and Blending Techniques
- Performance Chemicals
- Functional Chemicals

Module 5: Case Studies in Specialty Chemical Production

- Surfactants and Detergents
- Cosmetics and Personal Care Chemicals
- Perfumes and Flavouring Agents
- Industrial Applications and Green Chemistry
- Exercise: Develop the production of the following specialty chemicals:

Module 6: Case Studies in Fine Chemical Production

- Case Study: Production of Cyclopentanone (CPN) and Cyclopentyl Methyl Ether (CPME)
- Production Scheme (Simplified)
- Process Considerations

Module 7: Process Technology, Quality, and Sustainability

- Unit Operations and Process Control in Specialty/Fine Chemical Plants
- Quality Assurance and Control
- Safety, Health, and Environmental (SHE) Management

Assessment Components:

- Practical/Virtual Lab Reports (20%)
- Term Paper or Group Project (10%)
- Final Examination (70%)

Suggested Textbooks and References:

1. Pollak, P. – Fine Chemicals: The Industry and the Business
2. Winkler, H. – Production of Fine Chemicals
3. Anastas & Warner – Green Chemistry: Theory and Practice
4. Journal of Industrial and Engineering Chemistry
5. Organic Process Research & Development

Lecture Notes: Specialty and Fine Chemicals Production

Module 1.0 Introduction to Specialty and Fine Chemicals

1.1 Key Definitions

a) Fine Chemicals: Specific, Pure, single chemical substances produced in relatively small quantities through

complex synthesis, often used as intermediates or active ingredients in pharmaceuticals, agrochemicals, and

other high-value products. They are high-purity, complex molecules produced in small batches.

Often used as

building blocks or intermediates in pharmaceuticals and agrochemicals. Examples: Amino acids, drug

intermediates, peptide chains.

b) Specialty Chemicals: Also known as "performance chemicals." These are formulated products or mixtures

designed to provide specific functionalities and performance benefits rather than being pure chemical

substances. They are sold based on their function or performance rather than their chemical composition. They

are produced in moderate quantities, often as mixtures or blends, used in coatings, adhesives, cosmetics, and

industrial applications. They are low-volume, high-value chemical products. Often tailor-made for specific

performance or function. Examples: Paint additives, corrosion inhibitors, surfactants, biocides.

c) Commodity Chemicals: Also known as "bulk chemicals" or "basic chemicals." These are produced in very

large volumes (millions of tons annually), have relatively simple chemical structures, and are typically

undifferentiated by manufacturer. Their price is primarily determined by supply and demand. They are high-

volume, low-value chemicals, used as raw materials or intermediates in multiple sectors. Examples: Sulfuric

acid, ammonia, ethylene, sodium hydroxide.

Inter-relationships:

1) The value chain typically flows from commodities to fine chemicals or directly to specialty chemicals,

reflecting increasing complexity and value addition.

2) Commodity chemicals often serve as raw materials or basic building blocks for both fine and specialty

chemicals. (e.g., Benzene (commodity) used to synthesize intermediates for an API (fine chemical) or a

specific solvent (specialty chemical)).

3) Fine chemicals can be key ingredients within specialty chemical formulations (e.g., a high-purity biocide (fine chemical) used in a specialty paint (specialty chemical) for anti-fouling properties).

Diagram 1: Chemical Industry Classification

Bulk Chemicals (80%) vs. Fine/Specialty Chemicals (20%)

| |

└─Commodity (H₂SO₄, NH₃) └─High-value (Perfumes, APIs)

Example:

BASF's Verbund System: Integrates bulk + specialty chemical production for efficiency.

Diagram 2: Chemical Industry Value Chain

Raw Materials

Basic Chemicals (Commodities)

Intermediate Chemicals

→→→ Fine Chemicals (High purity)

→→→ Specialty Chemicals (Application-specific)

End-use Products (e.g., drugs, detergents, dyes)

1.2 Characteristics and Key Differences at a Glance

1.2.1 Commodity Chemical:

Characteristics:

- i. High Volume, Low Cost: Production is highly capital-intensive, relying on continuous processes to achieve economies of scale.
- ii. Low Value-Added: Profit margins per unit are typically low, requiring large quantities for profitability.
- iii. Standardized Quality: Products meet universal specifications (e.g., purity standards) but offer little differentiation.
- iv. Continuous Production: Manufacturing often involves large-scale, automated, continuous flow processes.

Examples:

- a) Acids: Sulfuric acid (H_2SO_4), Nitric acid (HNO_3), Hydrochloric acid (HCl).
- b) Bases: Sodium hydroxide (NaOH), Ammonia (NH_3).
- c) Salts: Sodium carbonate (Na_2CO_3), Chlorine (Cl_2), Sodium chloride (NaCl).
- d) Petrochemicals: Ethylene, Propylene, Benzene, Toluene, Xylene (BTX). These are foundational building blocks for polymers, plastics, and other organic chemicals.

1.2.2 Fine Chemicals

Characteristics:

- i. High Purity: Strict quality control and purification steps are essential, often requiring specific isomeric forms or very low impurity levels.
- ii. Low to Medium Volume, High Cost: Production is often batch-oriented due to lower volumes and specialized synthesis routes.
- iii. Complex Multi-step Synthesis: Involves intricate organic synthesis reactions, often with chiral centers, specific protecting groups, and careful control of reaction conditions.
- iv. High Value-Added: Significantly higher profit margins compared to commodity chemicals due to specialized knowledge, complex processes, and intellectual property.
- v. Market-driven: Production is often driven by specific customer demands or patent expiration of branded products.

Examples:

- a) Active Pharmaceutical Ingredients (APIs): The biologically active components of pharmaceutical drugs (e.g., paracetamol, ibuprofen, statins, specific anti-cancer agents).
- b) Advanced Intermediates: Complex molecules that are further processed to create APIs or other high-value products.
- c) Peptides and Oligonucleotides: Synthesized for therapeutic or research purposes.
- d) Specialty Catalysts: Highly selective catalysts used in specific industrial reactions.
- e) Biocides: Chemical substances or microorganisms that control harmful organisms (e.g., pesticides, antimicrobials for specific uses).

1.2.3 Specialty Chemicals

Characteristics:

- i. Performance-Driven: Designed to solve a specific problem or provide a unique attribute to an end product.
- ii. Functionality-Based: Value is in their effect, not just their chemical identity. Often a blend of several chemicals.
- iii. Niche Applications: Tailored for specific industries or applications, leading to smaller market volumes but very high margins.
- iv. Technical Service & R&D Intensive: Requires close collaboration with customers and significant R&D investment to develop customized solutions.
- v. Formulation Expertise: Success often lies in the art and science of blending and formulating components.

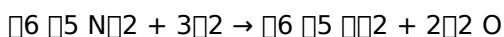
Examples:

- a) Adhesives and Sealants: Glues, epoxies, silicones for bonding materials.
- b) Paints, Coatings, and Inks: Formulations for protection, aesthetics, and printing.
- c) Surfactants: Used in detergents, personal care products, and industrial cleaners to modify surface tension.
- d) Lubricants and Fuel Additives: Enhance engine performance, reduce friction, or prevent corrosion.
- e) Agrochemicals: Formulated pesticides, herbicides, fungicides for specific crops and pests.
- f) Flavors and Fragrances: Complex mixtures that provide specific sensory experiences.
- g) Water Treatment Chemicals: Coagulants, disinfectants, corrosion inhibitors for water purification.
- h) Electronic Chemicals: High-purity chemicals used in the manufacturing of semiconductors, displays, and batteries (e.g., photoresists, etching chemicals).
- i) Plasticizers: Added to plastics to increase flexibility and workability.
- j) Enzymes: For industrial biotransformations.

Example 1: Aniline-Based Dyes

Used in textile and leather dyeing. Aniline is synthesized from nitrobenzene via catalytic hydrogenation.

Reaction:

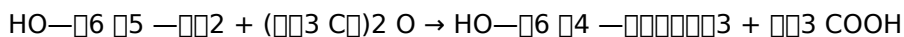
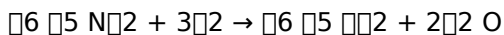


(Nitrobenzene) (Aniline)

Example 2: Paracetamol (Acetaminophen) – Fine Chemical

Used as an analgesic and antipyretic drug. Synthesized from p-aminophenol and acetic anhydride.

Reaction Diagram:



(p-aminophenol) (acetic anhydride) (Paracetamol)

1.2.4 Key differences

Feature Commodity Chemicals Specialty Chemicals Fine Chemicals

Very

Low to medium(Tons to Very low/small(Kilograms to

Volume/scale high/large(Millions and

hundreds of tons) hundreds of tons)

tons)

Single, high-purity

Product Type Bulk/basic Formulated mixtures

compounds

Unit Value Low Medium-High Very high

Customization No-Low Medium-High Very high

Purity Low-Medium Medium-High Very high

Formulations in large

scale depending on Blending, formulation,

Multistep organic synthesis,

factors like raw sometimes fermentation,

Production Methods biotechnological processes,

materials, market and chemical synthesis.

extraction, and purification.

demand, cost and

efficiency

Tailored to specific

Not tailored to specific customer or industry

Multipurpose batch plants

customers need but needs, often requiring

with strict quality control and

Facilities geared towards meeting unique properties such as

regulatory compliance.

the global market enhanced durability,

demand reactivity, or

performance

Batch or continuous,
 Production Mode Continuous Batch, multipurpose plants
 often formulated
 Global, undifferentiated Niche, specific industries Highly Niche, tailored
 Market
 (e.g., Pharma) solutions
 R &D Intensity Low-Medium Medium-High Very high
 Ammonia, H₂SO₄, Fragrances, polymers,
 Example Drug and agrochemical
 Sulfuric acid, ethylene, Dyes, Surfactants,
 intermediates, Active
 methanol surfactants, Food
 Feature Commodity Chemicals Specialty Chemicals Fine Chemicals
 additives, adhesives, pharmaceutical ingredients
 coatings (APIs)

1.3 Importance of Specialty and Fine Chemicals

Industry Relevance: Specialty and fine chemicals are critical to innovation in pharmaceuticals, materials

science, electronics, and more.

Skill Development: Prepares students for careers in R&D, quality assurance, and custom manufacturing

within the chemical industry

1.3.1 Industrial and Economic Significance

- Enable the functionality of high-tech industries (e.g., semiconductors, aerospace).
- Essential in consumer products: detergents, cosmetics, medicine.
- Drive innovation in targeted agriculture and personalized medicine.
- Contribute significantly to the global chemical economy despite smaller volumes.

1.3.2 Role in the Nigerian Economy

- Potential to reduce import dependence in agrochemicals, cosmetics, and pharmaceuticals.
- Supports the Nigerian Industrial Revolution Plan (NIRP).
- Opportunities in exploiting natural oils, herbs, and agricultural waste for fine chemical synthesis.

1.3.3 Economic Significance

- Key Economic Indicators: High revenue generation, significant employment, contribution to GDP, and substantial R&D investment. Global market: \$600+ billion (2023)
- Growth drivers: Bio-based chemicals, Asia-Pacific demand

The chemical industry is a cornerstone of the global economy, providing essential materials for virtually all

other manufacturing sectors (e.g., automotive, electronics, construction, healthcare, agriculture).

- Global Players: Dominated by multinational corporations, with significant production centers in North America, Europe, and Asia (especially China).

- Market Trends: Influenced by globalization, sustainability demands, technological advancements, and shifts in raw material availability.

Brief History and Evolution of the Chemical Industry

- Early Stages: Began with basic chemicals like sulfuric acid and soda ash (e.g., Leblanc process in the

late 18th century). Focus on large-scale production for textiles, glass, soap.

- 19th Century: Rise of organic chemistry, leading to synthetic dyes, pharmaceuticals, and explosives.

Emergence of the German chemical industry.

- 20th Century: Explosive growth with petrochemicals (from crude oil and natural gas), leading to polymers, fertilizers, and a vast array of synthetic materials. Development of modern process engineering.

- 21st Century: Increased focus on sustainability, green chemistry, biotechnology, digitalization, and specialization (fine and specialty chemicals).

1.4 Market Sectors for Specialty & Fine Chemicals

Sector Examples of Chemicals Used

Pharmaceuticals APIs, intermediates, chiral compounds

Agrochemicals Herbicides, insecticides, growth promoters

Electronics Photoresists, etching chemicals

Cosmetics Emollients, fragrances, UV blockers

Food & Beverages Flavourings, preservatives

Paints/Coatings Resins, dispersants, curing agents

1.5 Discussion Questions

1. Differentiate between specialty and commodity chemicals with examples.
2. Explain why the production of fine chemicals is mostly done in batches.
3. List three industries that rely heavily on specialty chemicals and explain why.
4. Using local Nigerian plants or materials, suggest one possible fine chemical that can be produced.

1.6 Summary

- Specialty and fine chemicals are crucial to innovation, offering performance-driven value in diverse sectors.

- They differ from commodity chemicals by volume, value, and application specificity.

- Nigeria has untapped potential for developing specialty chemical industries

using local raw materials.

Module 2.0 Classification and Functional Groups of Specialty Chemicals

2.1 Pharmaceuticals: These are high-purity, often complex organic molecules (APIs) or inactive substances (excipients) used in drug formulations.

Classifications/Chemical Classes:

- a) Active Pharmaceutical Ingredients (APIs): Small organic molecules (the majority of drugs), peptides, oligonucleotides, biologics (large molecules like proteins, antibodies).
- b) Drug Intermediates: Precursor molecules in the multi-step synthesis of APIs.
- c) Excipients: Inactive ingredients like binders (e.g., cellulose derivatives, povidone), fillers (e.g., lactose, microcrystalline cellulose), disintegrants (e.g., croscarmellose sodium), glidants (e.g., colloidal silicon dioxide), lubricants (e.g., magnesium stearate), coatings (e.g., hypromellose), solubilizers (e.g., polysorbates), preservatives (e.g., parabens), colorants, and flavoring agents.

Common Functional Groups (for APIs & Intermediates):

- i. Heterocyclic Compounds: Pyridines, pyrroles, imidazoles, indoles, quinolines, piperidines, furans, thiophenes, oxazoles, thiazoles. (Crucial for biological activity and binding).
- ii. Amines– common in APIs (e.g., antihistamines): Primary ($-NH_2$), secondary ($-NHR$), tertiary ($-NR_2$), quaternary ammonium salts ($-N^+R_4$). (Involved in biological interactions, often protonated at physiological pH).
- iii. Alcohols– in hydrophilic drugs and excipients ($-OH$): Hydroxyl groups for solubility, hydrogen bonding, and as synthetic handles.
- iv. Carboxylic Acids ($-COOH$): For salt formation, binding, and acidity. For solubility and reactivity
- v. Esters ($-COOR$) and Amides ($-CONHR$): Common linkages, involved in prodrug design or metabolic stability, found in many peptide-based drugs.
- vi. Ketones ($-C=O$) and Aldehydes ($-CHO$): Reactive carbonyls, often metabolic sites.
- vii. Halides ($-F$, $-Cl$, $-Br$, $-I$): Can modify lipophilicity, metabolic stability, or electronic properties.
- viii. Sulfonamides ($-SO_2NHR$), Sulfones ($-SO_2R$), Thioethers ($-SR$): Sulfur-containing functionalities.
- ix. Phenols ($Ar-OH$): Aromatic hydroxyls.

x. Common Functional Groups (for Excipients):

- xi. Hydroxyl ($-OH$): In polyols (e.g., mannitol, sorbitol), cellulose derivatives.
- xii. Carboxyl ($-COOH$): In cellulose derivatives (e.g., croscarmellose).
- xiii. Ether ($-OR$): In cellulose ethers (e.g., methylcellulose, HPMC).
- xiv. Esters ($-COOR$): In many lipid-based excipients, solubilizers.

Common Functional Groups (for Excipients):

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- d) Esters ($-COOR$): In many lipid-based excipients, solubilizers

Examples:

Ibuprofen (carboxylic acid)

Paracetamol (amide, phenol)

Lactose (excipient – hydroxyl groups)

2.2 Agrochemicals: Chemicals designed to protect crops by controlling pests. They often feature specific

functional groups that interact with biological targets.

Classification/Chemical Classes:

- i. Pesticides (General term): Includes herbicides, insecticides, and fungicides.
- ii. Organophosphates: Esters of phosphoric acid (e.g., malathion, chlorpyrifos).
- iii. Carbamates: Derivatives of carbamic acid (e.g., carbaryl).
- iv. Pyrethroids: Synthetic derivatives of natural pyrethrins (e.g., permethrin, cypermethrin).
- v. Sulfonylureas: Herbicides containing the sulfonylurea linkage (e.g., chlorsulfuron).
- vi. Triazines: Nitrogen-containing heterocyclic herbicides (e.g., atrazine).
- vii. Neonicotinoids: Insecticides structurally similar to nicotine (e.g., imidacloprid).
- viii. Strobilurins: Fungicides with a common β -methoxyacrylate group (e.g., azoxystrobin).
- ix. Glyphosate and related phosphonates: Herbicides.
- x. Triazoles: Fungicides (e.g., tebuconazole).

Common Functional Groups:

- a) Phosphate Esters: $(O=P(OR)_3)$
- b) Carbamates: $(-O-CO-NR_2)$
- c) Halogenated Hydrocarbons: (e.g., $-CCl_3$, $-CF_3$, $-CH_2Cl$)
- d) Nitriles $(-C\equiv N)$:
- e) Esters $(-COOR)$ and Amides $(-CONHR)$:
- f) Sulfonamides $(-SO_2NHR)$ and Sulfonylureas $(-SO_2NHCONH-)$:
- g) Heterocyclic rings: Triazines, imidazoles, pyrazoles, pyridines, furans, thiazoles.
- h) Aromatic rings: Often substituted with halogens, alkyl groups, nitro groups.
- i) Thioethers $(-SR)$ and Sulfoxides $(-S(O)R)$:
- j) Alcohols $(-OH)$ and Ethers $(-OR)$:

Examples:

Glyphosate (phosphonate, amine, carboxylate)

Atrazine (triazine ring)

Chlorpyrifos (organophosphate)

2.3 Flavours, fragrances, and food additives: These are compounds or mixtures designed to impart or

enhance specific sensory properties (taste, smell) or preserve and improve food quality.

Classification/Chemical Classes:

- i. Flavors: Esters, aldehydes, ketones, pyrazines, sulfur compounds, terpenes, lactones, acids.
- ii. Fragrances: Esters, alcohols, aldehydes, ketones, ethers, terpenes, macrocyclic musks.
- iii. Food Additives:
 - a. Preservatives: Organic acids (e.g., benzoic acid, sorbic acid and their salts), sulfites, nitrites.
 - b. Sweeteners: Polyols (e.g., xylitol, sorbitol), saccharides, artificial sweeteners (e.g., aspartame, sucralose, saccharin).

- c. Colorants: Azo dyes, carotenoids, anthocyanins.
- d. Antioxidants: Phenolic compounds (e.g., BHT, BHA, ascorbic acid).
- e. Emulsifiers: Mono- and diglycerides, lecithins, polysorbates.
- f. Thickeners/Stabilizers: Polysaccharides (e.g., guar gum, xanthan gum), modified celluloses.
- g. Acidity Regulators: Citric acid, phosphoric acid, sodium bicarbonate.

Common Functional Groups:

- a) Esters ($-\text{COOR}$): Key for many fruity (e.g., ethyl acetate, isoamyl acetate) and floral notes.
- b) Aldehydes ($-\text{CHO}$): Citrus (e.g., citral), green, fatty, and vanilla notes (e.g., vanillin).
- c) Ketones ($-\text{C}=\text{O}$): Fruity, creamy, woody notes (e.g., raspberry ketone, ionones).
- d) Alcohols ($-\text{OH}$): Floral (e.g., linalool, geraniol), woody, minty.
- e) Carboxylic Acids ($-\text{COOH}$): Acidity, tartness (e.g., citric acid, acetic acid).
- f) Phenols ($\text{Ar}-\text{OH}$): Spicy, smoky (e.g., eugenol, vanillin).
- g) Thiols ($-\text{SH}$) and Sulfides ($-\text{S}-$): Pungent, savory (e.g., in garlic, coffee).
- h) Lactones (cyclic esters): Creamy, fruity notes (e.g., δ -decalactone).
- i) Pyrazines, Pyrroles, Furans: Nitrogen/oxygen heterocycles for roasted, nutty, savory flavors.
- j) Ethers ($-\text{OR}$): Aniseed, camphor notes (e.g., anethole).
- k) Sulfonic Acids ($-\text{SO}_3\text{H}$) and their salts: In some synthetic food colorants.

Examples:

Vanillin (aldehyde, phenol, ether)

Citral (aldehyde, alkene)

Ascorbic acid (antioxidant – lactone, hydroxyl groups)

Eugenol (phenol, ether, alkene)

Methyl salicylate (wintergreen scent – ester, phenol)

2.4 Electronics: Extremely high-purity chemicals critical for manufacturing electronic components. Impurities at parts-per-billion or even parts-per-trillion levels can cause device failure.

Classification/Chemical Classes:

- i. Photoresists: Polymers containing photoactive compounds (PACs).
 - a. Positive Photoresists: Typically novolac resins with diazonaphthoquinone (DNQ).
 - b. Negative Photoresists: Often cyclized polyisoprene with bis-azide initiators, or chemically amplified resists (acid-catalyzed deprotection).
- ii. Developers: Solutions for patterning photoresists (e.g., Tetramethylammonium Hydroxide - TMAH).
- iii. High Purity Solvents: (e.g., propylene glycol monomethyl ether acetate - PGMEA, ethyl lactate, cyclohexanone).
- iv. Etchants: Highly pure acids (e.g., hydrofluoric acid, phosphoric acid, nitric acid) and bases, or gas mixtures for plasma etching.
- v. Dopants: Compounds containing elements like Boron, Phosphorus, Arsenic (e.g., diborane, phosphine, arsine) for altering semiconductor conductivity.

- vi. Precursors for CVD (Chemical Vapor Deposition) / ALD (Atomic Layer Deposition): Metal-organic compounds (e.g., trimethylaluminum, titanium tetraisopropoxide, silanes).
- vii. CMP (Chemical Mechanical Planarization) Slurries: Abrasive particles (e.g., silica, alumina) suspended in chemical solutions with polishing agents.

Common Functional Groups:

- a) Esters ($-\text{COOR}$): In many solvents (e.g., PGMEA, ethyl lactate), and as protecting groups in chemically amplified photoresists.
- b) Alcohols ($-\text{OH}$): In some solvents, polymer backbones.
- c) Phenols ($\text{Ar}-\text{OH}$): In novolac resins (photoresists).
- d) Diazo compounds ($-\text{N}=\text{N}-$): In photoactive compounds like DNQ.
- e) Epoxides (cyclic ethers): In some resin systems.
- f) Acrylates/Methacrylates: In many polymer resins for photoresists.
- g) Quaternary Ammonium Salts ($-\text{N}^+\text{R}_4$): In developers (e.g., TMAH).
- h) Metal-alkyl or Metal-alkoxide bonds: In CVD/ALD precursors (e.g., $-\text{Al}(\text{CH}_3)_3$, $-\text{Ti}(\text{OCH}(\text{CH}_3)_2)_4$).
- i) Silanes ($\text{R}_n\text{SiH}_{4-n}$) and Siloxanes (R_2SiO units): In silicon-containing polymers, precursors.

Examples:

Tetramethylammonium hydroxide (photoresist developer)

Diazonaphthoquinone (photoactive compound)

Hydrofluoric acid (etchant)

2.5 Personal care: Formulated chemicals for personal hygiene, beauty, and skin/hair/nail care, emphasizing safety, efficacy, and sensory attributes.

Classification/Chemical Classes:

- i. Surfactants: Anionic (e.g., Sodium Laureth Sulfate, Ammonium Lauryl Sulfate), cationic (e.g., Quaternary Ammonium Compounds like Cetrimonium Chloride), non-ionic (e.g., Polysorbates, fatty alcohol ethoxylates), amphoteric (e.g., Cocamidopropyl Betaine).
- ii. Emollients/Moisturizers: Fatty alcohols (e.g., cetyl alcohol), esters (e.g., isopropyl palmitate, caprylic/capric triglyceride), silicones (e.g., dimethicone), hydrocarbons (e.g., mineral oil, petrolatum), fatty acids.
- iii. Preservatives: Parabens (methylparaben, propylparaben), phenoxyethanol, organic acids (e.g., benzoic acid, sorbic acid), formaldehyde-releasing agents, isothiazolinones.
- iv. Thickeners/Rheology Modifiers: Polymers (e.g., carbomers, cellulose gums, xanthan gum), clays (e.g., bentonite).
- v. UV Filters: Organic (e.g., Avobenzone, Oxybenzone, Octinoxate) and inorganic (e.g., Zinc Oxide, Titanium Dioxide) UV absorbers.
- vi. Fragrances: (Refer to section 3).
- vii. Actives: Vitamins (e.g., Vitamin C - Ascorbic Acid, Vitamin E - Tocopherol), anti-aging compounds (e.g., retinoids - retinol, tretinoin; peptides), anti-acne agents (e.g., salicylic acid, benzoyl peroxide).
- viii. Conditioning Agents: Polyquaternium compounds, silicones, fatty amines.

Common Functional Groups:

- a) Sulfate ($-\text{OSO}_3^-$) and Sulfonate ($-\text{SO}_3^-$): In anionic surfactants.
- b) Carboxylates ($-\text{COO}^-$): In soaps (fatty acid salts) and some anionic surfactants/thickeners.
- c) Quaternary Ammonium Salts ($-\text{N}^+\text{R}_4$): In cationic surfactants, conditioning agents.
- d) Polyether chains ($-\text{O}-\text{CH}_2\text{CH}_2-$): In non-ionic surfactants, solubilizers.
- e) Esters ($-\text{COOR}$): In emollients, fragrances, some active ingredients.
- f) Alcohols ($-\text{OH}$) and Ethers ($-\text{OR}$): In solvents, emollients, polyols.
- g) Amides ($-\text{CONHR}$): In some surfactants (e.g., cocamide DEA) and thickeners.
- h) Silicones ($\text{Si}-\text{O}-\text{Si}$ backbone with organic substituents): In conditioning agents for skin and hair.
- i) Phenols ($\text{Ar}-\text{OH}$): In antioxidants, some preservatives, UV filters.
- j) Ketones ($-\text{C}=\text{O}$): In some fragrances.
- k) Aldehydes ($-\text{CHO}$): In some fragrances.

Examples:

Parabens (esters of para-hydroxybenzoic acid)

Cetyl alcohol (long-chain fatty alcohol)

Tocopherol (Vitamin E – phenolic antioxidant)

2.6 Industrial: Catalysts, corrosion inhibitors, Water treatment and functional Polymers, adhesives, and

sealants. These chemicals are used across various heavy and light industries to improve processes, protect assets, and create high-performance materials.

i. Classification/Chemical Classes (Catalysts):

- a. Heterogeneous Catalysts: Transition metals (e.g., Pt, Pd, Ni) on supports (e.g., alumina, silica, carbon), zeolites, metal oxides.
- b. Homogeneous Catalysts: Transition metal complexes (e.g., rhodium complexes for hydroformylation), organocatalysts, acid/base catalysts.
- c. Biocatalysts: Enzymes (e.g., lipases, oxidoreductases).

ii. Classification/Chemical Classes (Corrosion Inhibitors):

- a. Organic: Amines, phosphonates, imidazolines, carboxylic acids.
- b. Inorganic: Phosphates, nitrites, chromates (less common due to toxicity).

iii. Classification/Chemical Classes (Water Treatment):

- a. Coagulants/Flocculants: Inorganic salts (e.g., aluminum sulfate, ferric chloride), synthetic organic polymers (e.g., polyacrylamides, polydiallyldimethylammonium chloride - PolyDADMAC).
- b. Scale Inhibitors: Polycarboxylates, phosphonates, acrylates.
- c. Disinfectants: Chlorine-based compounds, ozone.
- d. pH Adjusters: Acids, bases.
- e. Biocides: Glutaraldehyde, quaternary ammonium compounds.

iv. Classification/Chemical Classes (Functional Polymers):

- a. Ion Exchange Resins: Cross-linked polystyrene with sulfonic acid or quaternary ammonium groups.
- b. Superabsorbent Polymers (SAPs): Cross-linked polyacrylates.
- c. Flocculants: High molecular weight polyacrylamides (anionic, cationic, non-ionic).

- d. Membrane Polymers: For filtration (e.g., polysulfones, PVDF).
- v. Classification/Chemical Classes (Adhesives & Sealants):
 - a. Polymer-based: Epoxies, polyurethanes, acrylics, silicones, hot melts (e.g., EVA, polyolefins), cyanoacrylates.
 - b. Natural Adhesives: Starch, casein.

Common Functional Groups:

- a) Amines ($-NH_2$, $-NHR$, $-NR_2$): In corrosion inhibitors, cationic flocculants, polyamines.
- b) Quaternary Ammonium Salts ($-N^+R_4$): In cationic flocculants, biocides, ion exchange resins.
- c) Carboxylic Acids ($-COOH$) and their salts ($-COO^-$): In polycarboxylates (scale inhibitors, dispersants), acrylic adhesives, some corrosion inhibitors.
- d) Sulfonic Acids ($-SO_3H$) and their salts ($-SO_3^-$): In some ion exchange resins, dispersants.
- e) Phosphonates ($R-PO(OH)_2$): In corrosion and scale inhibitors.
- f) Epoxides (cyclic ethers): In epoxy resins (adhesives).
- g) Urethanes ($-NH-CO-O-$): In polyurethane adhesives/sealants.
- h) Acrylate ($-CH=CHCOO-$) and Methacrylate ($-CH_2=C(CH_3)COO-$): In acrylic adhesives, SAPs, functional polymers.
- i) Siloxane ($Si-O-Si$) backbone: In silicone sealants.
- j) Amide ($-CONHR$): In polyacrylamides (flocculants).
- k) Hydroxyl ($-OH$): In polyols for polyurethanes.

Examples:

EDTA (chelating agent – amine, carboxylate)

Polyacrylamide (flocculant – amide group)

Bisphenol A diglycidyl ether (epoxy resin monomer)

2.7 Textiles: Dyes, pigments, and optical brighteners. These are chemicals used to color textile fibers or

enhance their whiteness and brightness.

i. Classification/Chemical Classes:

- a. Dyes (soluble): Azo dyes, anthraquinone dyes, phthalocyanine dyes, indigoid dyes, triphenylmethane dyes, reactive dyes, disperse dyes, basic dyes, acid dyes.
- b. Pigments (insoluble): Inorganic pigments (e.g., titanium dioxide, iron oxides), organic pigments (e.g., azo pigments, phthalocyanine pigments).
- c. Optical Brighteners (Fluorescent Whitening Agents - FWAs): Stilbene derivatives, coumarins, pyrazolines, naphthalimides.

ii. Common Functional Groups (for Dyes & Pigments):

- a. Chromophores: Groups responsible for color, typically extended conjugated systems (e.g., azo ($-N=N-$), carbonyl ($C=O$), nitro ($-NO_2$), quinoid structures, polyenes).
- b. Auxochromes: Groups that modify color intensity, hue, and aid solubility or binding to fibers (e.g., $-OH$, $-NH_2$, $-NR_2$, $-SO_3H$ (sulfonic acid), $-COOH$).
- c. Sulfonic Acids ($-SO_3H$) and Carboxylic Acids ($-COOH$): For water solubility and affinity to fibers

(especially acid and reactive dyes).

d. Amines ($-NH_2$, $-NHR$, $-NR_2$): Basic dyes, also auxochromes.

e. Hydroxyl ($-OH$) and Ether ($-OR$):

f. Reactive groups: For covalent bonding to fibers (e.g., vinyl sulfone, chlorotriazine in reactive dyes).

g. Aromatic and Heterocyclic Rings: Naphthalene, anthracene, quinoline, phthalocyanine systems.

Common Functional Groups (for Optical Brighteners):

a) Stilbene derivatives: Characterized by a $-CH=CH-$ conjugated system.

b) Sulfonic Acids ($-SO_3H$) and their salts: For water solubility and affinity to cellulose fibers.

c) Triazine rings: Often incorporated for structural stability and enhanced fluorescence.

d) Amine groups: For solubility and affinity.

e) Benzoxazole and Benzimidazole rings: Common heterocyclic systems in FWAs.

Examples:

Methyl orange (azo dye – azo and sulfonic acid groups)

Disperse Blue 56 (anthraquinone dye)

Fluorescent brighteners (stilbene derivatives)

Summary Table

Category	Examples	Common Functional Groups
Pharmaceuticals	APIs, excipients	Amides, amines, alcohols, acids, esters
Agrochemicals	Pesticides, fungicides	Organophosphates, carbamates, triazoles
Flavours, Fragrances & Vanillin	essential oils, Musk	Aldehydes, esters, phenols, terpenes, Ketones,
Food Additives	ketones, eugenol	lactones, phenols, ethers
Fragrances & Additives		
Electronics	Photoresists, etchants	Diazo, sulfonic acid, organometallics
Personal Care	Emollients, preservatives	Alcohols, esters, phenols, siloxanes
Industrial	Polymers, adhesives	Epoxides, isocyanates, acrylates, phosphonates
Textiles	Dyes, brighteners	Azo, sulfonic acid, anthraquinone, stilbenes

Module 3: Fine Chemicals Production

3.1: Fundamentals of Fine Chemical Synthesis

3.1.1 Reaction Mechanisms and Retrosynthesis in Fine Chemical Production

Reaction Mechanisms: Review of fundamental organic reaction mechanisms relevant to fine chemical

synthesis (e.g., nucleophilic substitution, electrophilic aromatic substitution, addition reactions, oxidation,

reduction). Understanding mechanisms is crucial for optimizing reactions, predicting side products, and

designing new routes.

Retrosynthesis: A powerful intellectual tool (pioneered by E.J. Corey) used to design synthetic routes. It

involves working backward from the target molecule to simpler, readily available starting materials.

Example: Retrosynthesis of Ibuprofen

Ibuprofen

(Target Molecule)

Disconnection: Break C-C bond between isobutyl and aromatic ring

Intermediate 1: Isobutyl group source

Intermediate 2: Aromatic ring with carboxylic acid

From isobutylbenzene and acetic acid derivative

Commercially available:

- Isobutylbenzene

- Acetyl chloride / Acetic acid

Alternative:

Ibuprofen

— Disconnection of C-C bond (between isobutyl and propanoic acid side chain)

Isobutylbenzene + α -Methylacrylic acid derivative (propanoic acid precursor)

| |

↓ ↓

Commercially available Commercially available

(petrochemical) (via Friedel-Crafts acylation)

3.1.2 Importance of Stereochemistry and Enantioselective Synthesis

Many fine chemicals, especially APIs, exhibit chirality (non-superimposable mirror images, enantiomers).

Enantiomers can have vastly different biological activities (e.g., one enantiomer is therapeutic, the other is

inactive or even toxic – recall the Thalidomide tragedy).

Stereochemistry: The study of the 3D arrangement of atoms in molecules.

Enantioselective Synthesis (Asymmetric Synthesis): Techniques used to produce a single enantiomer of a

chiral compound predominantly or exclusively.

Methods: Chiral auxiliaries, chiral catalysts (e.g., Sharpless epoxidation, Noyori hydrogenation), enzymes

(biocatalysis), chromatographic separation of racemates.

Example: Production of (S)-Ibuprofen, which is the active enantiomer, instead of a racemic mixture.

3.1.3 Catalysis in Fine Chemical Synthesis

Catalysts are critical for enhancing reaction rates, improving selectivity (chemo-, regio-, stereo-), and

reducing energy consumption.

Homogeneous Catalysis: Catalyst is in the same phase as reactants (e.g., metal complexes like Wilkinson's

catalyst for hydrogenation).

Advantages: High selectivity, good reproducibility.

Disadvantages: Separation from products can be difficult.

Heterogeneous Catalysis: Catalyst is in a different phase (usually solid catalyst, liquid/gas reactants) (e.g.,

palladium on carbon for hydrogenation, Ziegler-Natta catalysts).

Advantages: Easy separation.

Disadvantages: Lower selectivity often, active site heterogeneity.

Biocatalysis (Enzymatic Synthesis): Using enzymes as catalysts.

Advantages: High specificity and selectivity (especially enantioselectivity), mild reaction conditions (aqueous, ambient temperature/pressure), environmentally friendly.

Disadvantages: Enzyme stability, cost, narrow substrate scope for some enzymes.

Example: Lipases for kinetic resolution of racemic mixtures, oxidoreductases for specific oxidations/reductions.

Comparison Table

Catalyst Reactant Interaction Schematic

Catalysis Type Example

Phase Phase Description Representation

Catalyst and

reactants are in the

Same as

same phase, $\text{C} + \text{R} \rightarrow \text{C}\cdots\text{R}$ (mixed) Acid-catalyzed

reactants Liquid or (Catalyst and

Homogeneous allowing uniform esterification using

(usually gas reactants in one liquid

mixing and H_2SO_4

liquid) phase)

molecular-level

interaction.

Catalyst is in a

different phase (e.g., $\text{C}_{\text{solid}} + \text{R} \rightarrow \text{C}\cdots\text{R}$

Different Liquid or solid in liquid); surface interaction Hydrogenation

Heterogeneous

(usually solid) gas reactants must (Reactants adsorb on using Pd/C

diffuse to catalyst solid catalyst surface)

surface.

Catalyst Reactant Interaction Schematic

Catalysis Type Example

Phase Phase Description Representation

Enzyme or microbe

Aqueous binds reactant at an $\text{C} + \text{R} \rightarrow \text{C}\cdots\text{R}$ (lock-

Liquid or

enzyme or active site; highly and-key fit) (Enzyme Lipase-catalyzed

Biocatalysis biological

whole-cell specific and operates binds specific transesterification

media

(biological) under mild substrate) conditions.

Key Legend:

□ ◆ = molecular catalyst

□ □ = solid catalyst surface

□ □ = liquid reactant

□ ◆ = enzyme or biological catalyst

3.1.4 Multi-step Synthesis Strategies and Reaction Optimization

Most fine chemicals require a sequence of multiple reactions. Strategic planning is crucial.

Linear vs. Convergent Synthesis:

Linear: $A + B \rightarrow C$; $C + D \rightarrow E$. Builds the molecule sequentially. Less efficient if intermediate steps fail.

Convergent: $A + B \rightarrow C$; $D + E \rightarrow F$; $C + F \rightarrow G$. Synthesizes fragments independently and then joins them.

More efficient, especially for complex molecules.

Reaction Optimization: Adjusting reaction parameters to maximize yield, purity, selectivity, and minimize

side products and waste.

Parameters: Temperature, pressure, solvent, catalyst loading, reactant ratios, stirring speed, reaction time.

Techniques: Design of Experiments (DoE), high-throughput screening, process analytical technology (PAT).

3.2 Production of Active Pharmaceutical Ingredients (APIs)

3.2.1 Drug Discovery and Development Pipeline

Phases:

1. Target Identification & Validation: Identifying a biological target (e.g., enzyme, receptor) involved in a disease.

2. Lead Discovery & Optimization: Finding small molecules (leads) that interact with the target, then optimizing their potency, selectivity, and pharmacokinetic properties.

3. Pre-clinical Development: In vitro (test tube) and in vivo (animal) testing for efficacy, toxicity, and pharmacokinetics.

4. Clinical Trials (Phase I, II, III): Human testing for safety, dosage, efficacy, and side effects.

5. Regulatory Approval: Submission of a New Drug Application (NDA) to regulatory bodies (e.g., FDA in USA, NAFDAC in Nigeria).

6. Post-Market Surveillance: Continued monitoring after drug approval.

Role of APIs: The output of the synthesis phase, providing the active component for the drug product.

3.1: Drug Development Pipeline (A linear or circular diagram showing the sequential stages from "Target

ID" to "Marketed Drug," indicating the iterative nature and high attrition rate)

3.2.2 Batch vs. Continuous Manufacturing Processes for APIs

Batch Manufacturing:

Description: Reactants are loaded into a reactor, reaction proceeds, product is isolated, then the reactor is

cleaned for the next batch.

Advantages: Flexible for multi-product facilities, easier to handle different reaction conditions for different

steps, established regulatory framework.

Disadvantages: Lower productivity per unit volume, higher labor cost, more difficult to achieve consistent

product quality, longer cycle times, higher energy consumption (due to heating/cooling cycles).

Continuous Manufacturing:

Description: Reactants are continuously fed into a reactor, and products are continuously withdrawn.

Reactions occur in flow reactors (e.g., microreactors, plug flow reactors).

Advantages: Higher productivity, smaller footprint, better quality control (steady-state operation), lower

energy and solvent use, safer (smaller reaction volumes).

Disadvantages: High initial capital investment, less flexible for product changeovers, complex process

control, evolving regulatory landscape.

Trend: Increasing adoption, especially for high-volume APIs, due to efficiency and quality benefits.

3.2: Batch vs. Continuous Reactor Setups(Simple diagrams showing a stirred tank reactor for batch vs. a

series of smaller flow reactors for continuous processing)

3.2.3 Good Manufacturing Practices (GMP) and Regulatory Compliance

GMP: A system of quality control guidelines and regulations to ensure that products are consistently produced and controlled according to quality standards.

Key Principles: Quality Management System, Qualified Personnel, Adequate Facilities & Equipment, Validated Processes, Documented Procedures (SOPs), Effective Record Keeping, Complaint Handling,

Recall Procedures.

Regulatory Bodies:

NAFDAC (National Agency for Food and Drug Administration and Control) in Nigeria.

FDA (Food and Drug Administration) in the USA.

EMA (European Medicines Agency) in Europe.

Importance: Ensures safety, efficacy, and quality of pharmaceutical products, protecting public health. Non-

compliance can lead to severe penalties, product recalls, and market bans.

3.2.4 Scale-Up Challenges and Process Validation

Scale-Up: The process of transitioning a chemical reaction or process from laboratory scale to pilot plant

scale and then to full industrial production.

Challenges: Heat transfer (exothermic reactions can become uncontrollable), mixing efficiency, mass

transfer, solvent selection, impurity profile changes, material handling, crystallization behavior.

Example: An exothermic reaction that is easily controlled in a small flask may become dangerous in a large

reactor due to reduced surface area-to-volume ratio, leading to poor heat dissipation and potential runaway

reactions.

Process Validation: Documented evidence that a process consistently produces a product meeting its pre-

determined specifications and quality attributes.

Phases: Process design, process qualification, continued process verification.

Importance: Required by regulatory authorities to ensure product consistency and quality.

3.3 Other Fine Chemical Categories

3.3.1 Agrochemicals: Synthesis of Herbicides, Insecticides, Fungicides

Definition: Chemicals used in agriculture to protect crops from pests (insects, weeds, fungi) and diseases,

enhancing crop yield and quality. They are highly specialized fine chemicals.

Synthesis: Often involves complex organic synthesis routes to create highly selective and effective molecules.

a. Herbicides: Kill or control unwanted plants (weeds). (e.g., Glyphosate – inhibits an enzyme involved

in amino acid synthesis in plants).

b. Insecticides: Control insect pests. (e.g., Pyrethroids – synthetic versions of natural pyrethrins, affecting insect nervous systems).

c. Fungicides: Control fungal diseases in plants. (e.g., Triazoles).

Challenges: Environmental impact, toxicity, development of resistance, regulatory scrutiny, demand for

"greener" alternatives.

3.3.2 Intermediates for Polymers and Materials

Definition: Specialized monomers and chemical additives that impart specific properties to polymers and

advanced materials (e.g., flame retardants, UV stabilizers, antioxidants, cross-linking agents, specialized

initiators).

Synthesis: Often requires high purity and specific isomeric forms to ensure optimal performance in the final material.

Example: Synthesis of specific diisocyanates for polyurethanes with tailored properties, or specialized

catalysts for stereospecific polymerization.

3.3.3 Specialty Solvents and Reagents

Definition: High-purity solvents and reagents used in specific laboratory syntheses, analytical methods, or

industrial processes where standard commodity solvents are insufficient.

Characteristics: Extremely low impurity levels, specific water content, specific spectroscopic purity, often

anhydrous or highly purified for sensitive reactions.

Manufacturing: Involves rigorous purification techniques (e.g., advanced distillation, adsorption, inert

atmosphere handling) and stringent quality control.

Examples: Anhydrous THF for organometallic reactions, deuterated solvents for NMR spectroscopy, high-

purity acids and bases for trace analysis.

3.3.4 Biocides: Synthesis and Application

Definition: Chemical substances or microorganisms intended to destroy, deter, render harmless, prevent the

action of, or otherwise exert a controlling effect on any harmful organism (e.g., bacteria, fungi, algae, pests).

Applications: Water treatment (disinfectants), paints (anti-fouling agents), wood preservatives, textiles,

personal care products (preservatives), agricultural uses (fungicides, herbicides, insecticides as a broader

category).

Synthesis: Diverse chemical structures, often involving complex organic synthesis.

Regulation: Highly regulated due to potential toxicity and environmental impact (e.g., requiring approval

from environmental protection agencies).

Module 4: Specialty Chemicals Production and Applications

4.1 Formulation and Blending Techniques

4.1.1 Principles of Formulating Multi-component Mixtures

Definition: Formulation is the process of combining different chemical substances to create a final product

with specific performance characteristics. Unlike fine chemicals, the end product is often a mixture, not a

single pure compound.

Key Principles:

Functionality: Each component contributes to the desired performance (e.g., a binder for adhesion, a pigment

for color, a rheology modifier for flow).

Compatibility: Components must be physically and chemically compatible to ensure stability, prevent

undesirable reactions, and maintain performance over time.

Synergy: Components can interact to produce a combined effect greater than the sum of their individual

effects.

Cost-Effectiveness: Balancing performance requirements with raw material costs and manufacturing efficiency.

Safety and Environmental Impact: Ensuring the final formulation is safe for use and environmentally responsible.

4.1.2 Methods: Dissolution, Neutralization, Dispersion, Emulsification, Blending

Dissolution: Dissolving solid components into a liquid solvent. Important for clear solutions and uniform distribution.

Neutralization: Adjusting pH by adding acids or bases to achieve optimal stability or performance (e.g., in detergents, personal care products).

Dispersion: Distributing fine solid particles evenly throughout a liquid medium (e.g., pigments in paint).

Requires high-shear mixing equipment.

Emulsification: Creating a stable mixture of two immiscible liquids (e.g., oil-in-water or water-in-oil emulsions). Requires surfactants (emulsifiers) and high-shear mixing.

4.1: Emulsion Types (Oil-in-Water vs. Water-in-Oil)(Simple diagrams showing droplets of one liquid dispersed in another continuous phase, stabilized by surfactant molecules at the interface:Blending: Simple

mixing of liquids or powders to achieve homogeneity. Can range from simple agitation to complex high-shear

blending. Granulation: Forming powders into larger granules, often for ease of handling or controlled release)

4.1.3 Role of Excipients and Additives in Achieving Desired Performance

Excipients: Generally used in pharmaceuticals or food. Inactive ingredients used as carriers, binders, fillers, disintegrants, etc., to aid drug delivery or food processing.

Additives (General): Chemicals added in small quantities to a base formulation to modify its properties or enhance performance.

Examples:

Rheology Modifiers (Thickeners): Control viscosity and flow properties (e.g., in paints, cosmetics).

Dispersants: Prevent aggregation of particles in a suspension.

Wetting Agents: Improve the spread of a liquid over a surface.

UV Stabilizers: Protect products from degradation by UV radiation.

Antioxidants: Prevent oxidation and extend shelf life.

Preservatives: Prevent microbial growth.

Dyes and Pigments: Provide color.

4.2 Performance Chemicals

4.2.1 Adhesives and Sealants

Definition:

Adhesives: Substances capable of holding materials together by surface attachment (adhesion) and internal

strength (cohesion).

Sealants: Materials used to prevent fluid leakage or penetration by filling gaps between substrates. They

maintain sealing properties even when there is joint movement.

Chemistry: Based on various polymer types: epoxies, polyurethanes, acrylics, silicones, hot melts, cyanoacrylates.

Manufacturing: Blending of polymers, resins, hardeners, fillers, and additives.

Applications: Construction, automotive, aerospace, packaging, electronics, medical devices.

4.2.2 Paints, Coatings, and Inks

Definition: Complex formulations designed to protect, decorate, or mark surfaces.

Components:

Pigments: Provide color, opacity, and sometimes protection (e.g., TiO_2 for white, iron oxides for reds/browns).

Binders (Resins): The film-forming component that holds the pigment particles together and adheres the

paint to the surface (e.g., acrylics, alkyds, epoxies).

Solvents: Dissolve the binder and provide viscosity for application; evaporate after application (e.g., water,

hydrocarbons).

Additives: Small amounts of chemicals to improve properties (e.g., rheology modifiers, dispersants, anti-

foaming agents, UV stabilizers).

Manufacturing: Dispersion of pigments in binders, mixing, grinding (milling), and final blending of all

components.

Applications: Decorative (walls), protective (corrosion prevention on metals), functional (anti-fouling marine

coatings, conductive inks).

4.2: Cross-section of a Painted Surface (Illustrates the layers: substrate -> primer -> paint/coating film

(showing pigments embedded in a binder matrix).

4.2.3 Surfactants and Detergents

Definition:

Surfactants (Surface Active Agents): Molecules with both hydrophilic (water-loving) and hydrophobic

(water-fearing) parts. They reduce surface tension between two liquids, a liquid and a gas, or a liquid and a

solid.

Detergents: Formulations containing surfactants as their primary active ingredients, designed for cleaning purposes.

Types of Surfactants: Anionic, cationic, non-ionic, amphoteric (based on the charge of their hydrophilic head group).

Synthesis: Primarily from petrochemicals (e.g., ethylene oxide for non-ionics) or oleochemicals (from natural oils/fats for soaps and some detergents).

Industrial Production: Complex synthesis and blending processes, often involving ethoxylation, sulfonation, or esterification.

Applications: Cleaning products (laundry, dishwashing), personal care (shampoos, soaps), emulsifiers in food, pharmaceuticals, agriculture, oil & gas recovery.

4.3: Micelle Formation by Surfactants (Shows surfactant molecules forming a spherical micelle in water, with hydrophobic tails oriented inwards and hydrophilic heads outwards, encapsulating a grease droplet.

4.2.4 Lubricants and Fuel Additives

Definition:

Lubricants: Substances introduced between two moving surfaces to reduce friction, wear, and heat generation. (e.g., engine oils, greases).

Fuel Additives: Chemicals added to fuels in small quantities to improve performance, enhance combustion, reduce emissions, or protect engine components.

Composition:

Lubricants: Base oils (mineral, synthetic, or semi-synthetic) and various performance additives (e.g., dispersants, detergents, anti-wear agents, viscosity index improvers, antioxidants).

Manufacturing: Blending of base oils with specific additive packages to meet performance specifications.

Applications: Automotive, industrial machinery, aerospace, marine.

Fuel Additives Examples: Octane improvers, detergents (to prevent deposits), cold flow improvers (for diesel), corrosion inhibitors.

4.3 Functional Chemicals

4.3.1 Flavors and Fragrances

Definition: Complex mixtures of aromatic compounds designed to impart specific tastes (flavors) or smells (fragrances) to products.

Sources:

Natural: Extraction from plants (e.g., essential oils from flowers, fruits), animal sources.

Synthetic: Chemical synthesis of specific aroma compounds, often derived from petrochemicals or terpenes.

Manufacturing:

Extraction: Distillation, solvent extraction, enfleurage for natural products.

Synthesis: Multi-step organic synthesis for synthetic compounds.

Blending/Formulation: The art and science of combining individual aroma chemicals and extracts to create

unique and complex profiles. Requires skilled perfumers/flavorists.

Applications: Food and beverages, cosmetics, personal care products, detergents, pharmaceuticals, industrial

products.

4.3.2 Food Additives

Definition: Substances added to food to preserve flavor or enhance taste, appearance, or other qualities, often

during processing or storage.

Types and Applications:

Preservatives: Prevent spoilage by microbial growth or oxidation (e.g., sodium benzoate, citric acid, nitrates).

Sweeteners: Provide sweetness with fewer calories (e.g., aspartame, saccharin, sucralose).

Colorants: Impart specific colors (natural or synthetic dyes).

Emulsifiers, Thickeners, Stabilizers: Improve texture and consistency.

Antioxidants: Prevent oxidative rancidity (e.g., BHT, BHA).

Production: Chemical synthesis, fermentation (for some enzymes or flavors), extraction from natural

sources.

Regulation: Highly regulated by food safety authorities (e.g., NAFDAC, FDA) due to direct human consumption.

4.3.3 Water and Waste Treatment Chemicals

Definition: Chemicals used to purify water (drinking water, wastewater, industrial process water) and treat

industrial waste streams to make them safe for discharge or reuse.

Types and Functions:

Coagulants/Flocculants: Promote aggregation of suspended particles for easier removal (e.g., aluminum

sulfate, ferric chloride, synthetic polymers).

Disinfectants: Kill pathogens (e.g., chlorine, ozone, UV, chlorine dioxide).

pH Adjusters: Acids or bases to optimize treatment processes.

Corrosion Inhibitors: Prevent corrosion in water systems.

Scale Inhibitors: Prevent mineral scale formation.

Activated Carbon: For adsorption of organic pollutants.

Manufacturing: Large-scale chemical production for basic chemicals (e.g., chlorine, alum) and specialized

synthesis for advanced polymers or adsorbents.

Applications: Municipal water treatment plants, industrial wastewater treatment, cooling towers, boiler water treatment.

4.3.4 Electronic Chemicals

Definition: High-purity chemicals specifically manufactured for use in the electronics industry, particularly

for semiconductor manufacturing, displays, and advanced batteries.

Characteristics: Extremely high purity (often ppb or ppt levels), low metal content, specific particle size

distribution, very tight quality specifications. Impurities can severely impact device performance.

Types:

Photoresists: Light-sensitive polymers used in photolithography to define circuit patterns.

Etchants: Chemicals used to selectively remove material (e.g., acids, bases, plasma etching gases).

Cleaners: High-purity solvents for removing residues.

Dopants: Chemicals used to alter the electrical properties of semiconductors.

Precursors: For chemical vapor deposition (CVD) or atomic layer deposition (ALD).

Production: Involves highly controlled environments (cleanrooms), specialized purification techniques, and

rigorous analytical testing.

Applications: Microchips, integrated circuits, flat-panel displays, LEDs, solar cells, advanced battery components.

Module 5: Case Studies in Specialty Chemical Production

5.1 Surfactants and Detergents

Types of surfactants (anionic, cationic, non-ionic, amphoteric)

Industrial production (sulfonation, ethoxylation, esterification)

Applications in detergents, personal care, and oil recovery

Demo: Foam stability test (comparing different surfactants)

Problem-solving: Calculate molar ratios for ethoxylation

Assessment: Lab report on surfactant properties

5.2 Cosmetics and Personal Care Chemicals

Key ingredients (emollients, preservatives, thickeners)

Manufacturing processes (hydrogenation, encapsulation)

Regulatory standards (FDA, EU Cosmetics Regulation)

Assessment: Formulating a lotion (ingredient selection), (design a shampoo with specified properties)

Synthetic vs. Natural Ingredients in Cosmetics"

5.3 Perfumes and Flavouring Agents

Natural vs. synthetic fragrances

Extraction methods (steam distillation, CO₂ extraction)

Industrial synthesis of vanillin, menthol, and musks

Assessment: Lab Work: Essential oil extraction from orange peel. Practical report on essential oil yield, essay

on sustainability in fragrance production

5.4 Industrial Applications and Green Chemistry

Bio-based surfactants (sugar-based, amino acid-derived)

Waste management in chemical production

Carbon footprint reduction strategies

Design a "green" detergent

Presentation on bio-alternatives in industry

5.4.1 Traditional Black Soap Preparation (Local Nigerian Style)

Black soap is cherished across West Africa, especially Nigeria and Ghana, for its rich, natural skin-care benefits.

Introduction

African black soap, or simply black soap (also known by various local names such as *ọṣe dúdú*, *sabulun salo*, and *ncha*

nkota), is a kind of soap originating in Nigeria, invented by the Yoruba people. It is made from the ash of various burnt

organic materials such as cocoa pods, woods, palm kernel bunch and plantain leaves.

The steps involved include:

- Ash is mixed with water in a stainless or clay pot and stirred until it forms paste.
- It is then left to rest for about 3 days (3 days for best quality).
- The ash sources include burnt cocoa pods, banana leaves/stems, palm kernel bunch (*isi akwu*).
- After resting, the ash water mixture is filtered using a 0.1mm mesh and a brown-colour filtrate is obtained.
- This filtrate is boiled for six hours or more to sterile it and get a concentrated lye solution.
- The boiled water contain sodium hydroxide which is necessary for saponification (turning oil to soap). The

longer it boils, the better the lie quality.

- The lye solution is ready when it has darkened.
- Additives including Blue vitriol, alum, sulphur and trona are then added at concentration of 400mg/ml of lye solution.

5.1: Pictures of waste Palm kernel bunch

5.2: Picture of waste cocoa pods

- Palm kernel oil is then poured in a pot and heated briefly for 5 minutes.
- Then the lye is mixed with pal kernel oil (5:1). Palm kernel oil is the best (it provides hydration while the lye exfoliates).
- The type of oil used and the lye:oil ratio is what makes the difference between soaps
- Then continual stirring the lye:oil mixture with regulation of the heat supply to avoid burning the soap or igniting the solution.

- The lye should be added bit by bit into the oil to avoid “lye volcano” which is very dangerous.
- Ensure all the lye are thoroughly mixed with all the oil to ensure complete saponification.
- Halt cooking when the mixture hardens (when it is cohesive and unified in nature).
- To achieve the desired texture and appearance, the soap is pounded until it becomes smooth and pliable.
- Its then gently heated to harden more and molded into shapes for use.

You can leverage on the information given plus the fact that the Country is in a very poor economic state, and search for

more effective, cheaper and locally-sourced alternative raw materials to feed our industries is inevitable.

5.4.2 Production of Acrybon

Introduction

- Acrybond is a high quality water-based adhesive that has multiple usage.
- It is used for fastening materials , this include bonding of floor tiles, wood rubber, formica etc.
- Similar product in the market is topbond

Materials Needed

- (1) Acrylic resin
- (2) Water
- (3) Nitrosol
- (4) Ammonia

Functions

- (1) Acrylic resin: Possesses adhesive properties
- (2) Water: Act as the solvent
- (3) Nitrosol: Thickener
- (4) Ammonia: Neutraliser

Estimated for the production of 270 g acrybond

- (1) Acrylic resin: 205.3g
- (2) Water: 68.4g
- (3) Nitrosol: 2.7g
- (4) Ammonia: 1.4g

Production procedure

- (1) Pour 205.3 g of Acrylic resin into a container
- (2) Add 68.4g of water and stir for 3 minutes
- (3) Add 2.7g of Nitrosol and stir for 5 minutes
- (4) Add 1.4g of Ammonia and stir for 3 minutes

The gum is now ready for use.

5.5 Exercise: Develop the production of the following specialty chemicals:

Paracetamol (acetaminophen) synthesis

Aniline dyes and intermediates

Citronellal and other fragrance precursors

Chlorpyrifos and organophosphates

Vitamin C and nutraceuticals

Analysis of process efficiency and challenges

Module 6: Case Studies in Fine Chemical Production

6.1. Case Study: Production of Cyclopentanone (CPN) and Cyclopentyl Methyl Ether (CPME)

Overview

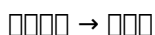
□ Cyclopentanone (CPN) and Cyclopentyl Methyl Ether (CPME) are valuable fine chemicals used as solvents and intermediates.

□ Traditional production involves multi-step reactions starting from dicyclopentadiene (DCPD).

6.2 Production Scheme (Simplified)

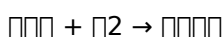
i. Thermal cracking of Dicyclopentadiene (DCPD) to Cyclopentadiene (CPD):

heat



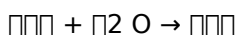
ii. Selective Hydrogenation of CPD to Cyclopentene (CPE):

Ni catalyst



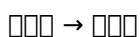
iii. Hydration of CPE to Cyclopentanol (CPL):

Acid catalyst



iv. Dehydrogenation of CPL to Cyclopentanone (CPN):

Dehydrogenation catalyst



6.3 Process Considerations

□ The batch hydrogenation of CPD requires careful temperature control to avoid runaway polymerization.

□ Gas phase continuous hydrogenation methods offer safety advantages but are less commercially implemented.

□ Purification of CPN is energy-intensive due to azeotropic mixtures with water.

Process Flow Diagram (Block Diagram) for Cyclopentanone Production

Step Equipment/Operation Purpose

1. Thermal Cracking Cracking furnace Convert DCPD to CPD
2. Selective Hydrogenation Batch reactor with Ni catalyst Convert CPD to CPE
3. Hydration Acid-catalyzed reactor Convert CPE to CPL
4. Dehydrogenation Dehydrogenation reactor Convert CPL to CPN
5. Separation & Purification Distillation columns Purify CPN from byproducts

[DCPD Feed] --> [Cracking Furnace] --> [Selective Hydrogenation Reactor] --> [Hydration Reactor] -->

[Dehydrogenation Reactor] --> [Distillation & Purification] --> [CPN Product]

6.1: Block flow diagram of Cyclopentanone production process

Module 7: Process Technology, Quality, and Sustainability

7.1 Unit Operations and Process Control in Specialty/Fine Chemical Plants

7.1.1 Key Unit Operations:

Reaction Engineering: Designing and operating reactors (batch, semi-batch, continuous flow) to optimize

reaction conditions.

Separation Processes:

Distillation: Used for purifying volatile components, solvent recovery. High-efficiency columns for fine

chemicals.

Filtration: Separating solids from liquids (e.g., product isolation, impurity removal). Various types: pressure

filters, centrifuges, membrane filtration.

Crystallization: Purifying solid products by forming crystals from solution. Crucial for purity, particle size,

and morphology control.

Extraction: Separating components between two immiscible liquid phases.

Drying: Removing solvent from solid products to meet specifications.

Mixing and Blending: Ensuring homogeneity for reactions and formulations (various mixer types: agitators,

homogenizers, high-shear mixers).

Material Handling: Safe and efficient transfer of raw materials, intermediates, and products (solids, liquids,

gases).

7.1.2 Process Analytical Technology (PAT) and Automation

PAT: A system for designing, analyzing, and controlling manufacturing processes through timely measurements of critical quality and performance attributes of raw and in-process materials and processes

with the goal of ensuring final product quality.

Techniques: In-line or at-line spectroscopic methods (e.g., NIR, FTIR, Raman), chromatography (e.g.,

HPLC), particle size analysis.

Benefits: Real-time monitoring, tighter process control, reduced batch failures, faster release of products.

Automation: Using automated systems (e.g., DCS - Distributed Control Systems, PLCs - Programmable

Logic Controllers) to control and monitor chemical processes.

Benefits: Improved safety, consistency, efficiency, reduced labor costs, precise control over reaction

parameters.

7.1.3 Reactor Design Considerations for Batch and Semi-Batch Processes

Batch Reactors: Commonly used for fine chemicals due to smaller volumes, flexibility for multi-product

lines, and easier handling of complex, multi-step syntheses.

Considerations: Heat transfer (heating/cooling jackets, internal coils), agitation (mixing efficiency), material

compatibility, pressure control, charging/discharging mechanisms.

Semi-Batch Reactors: One reactant is added gradually to another.

Advantages: Better control of exothermic reactions, improved selectivity by maintaining low concentration

of a reactive intermediate.

Design Focus: For fine/specialty chemicals, reactor design emphasizes precision, safety, flexibility, and

containment rather than just maximum throughput.

7.2 Quality Assurance and Control

7.2.1 Analytical Methods for Purity and Characterization

Chromatography:

HPLC (High-Performance Liquid Chromatography): For separation and quantification of non-volatile or

thermally labile compounds. Crucial for API purity.

GC (Gas Chromatography): For volatile compounds, purity checks, solvent residue analysis.

Spectroscopy:

NMR (Nuclear Magnetic Resonance): For structural elucidation and quantitative analysis.

FTIR (Fourier-Transform Infrared Spectroscopy): For functional group identification and reaction monitoring.

UV-Vis Spectroscopy: For quantification, purity, and reaction progress.

Mass Spectrometry (MS): For molecular weight determination, impurity identification, and structural analysis (often coupled with GC or HPLC).

Titration: For quantitative analysis of specific components.

Physical Property Testing: Melting point, boiling point, refractive index, density, viscosity.

7.2.2 Statistical Process Control (SPC)

Definition: A method of quality control that uses statistical methods to monitor and control a process. It helps

ensure that a process operates efficiently, producing more specification-conforming product with less waste.

Control Charts: Visual tools used to monitor process variation over time. Data points are plotted and

compared to control limits.

Importance: Helps identify "out of control" conditions, allowing corrective action before defects occur,

improving consistency and quality.

7.2.3 Quality Management Systems (e.g., ISO 9001)

Definition: A formalized system that documents processes, procedures, and responsibilities for achieving

quality policies and objectives.

ISO 9001: An internationally recognized standard for quality management systems (QMS).

Key Principles: Customer focus, leadership, engagement of people, process approach, improvement,

evidence-based decision making, relationship management.

Benefits: Improved product quality, increased customer satisfaction, enhanced efficiency, better regulatory

compliance, improved market reputation.

7.3 Safety, Health, and Environmental (SHE) Management

7.3.1 Hazard Identification and Risk Assessment

Chemical production involves inherent hazards (flammability, toxicity, corrosivity, explosivity).

Hazard Identification: Recognizing potential sources of harm (e.g., reactive chemicals, high pressures/temperatures, dangerous equipment).

Risk Assessment: Evaluating the likelihood and severity of harm from identified hazards.

Tools: HAZOP (Hazard and Operability study), FMEA (Failure Mode and Effects Analysis), Job Safety Analysis (JSA).

Risk Mitigation: Implementing control measures (e.g., engineering controls - ventilation, containment;

administrative controls - procedures, training; PPE - Personal Protective Equipment).

7.3.2 Waste Minimization, Treatment, and Disposal

Chemical processes often generate significant waste streams (solvents, by-products, unreacted raw materials).

Waste Minimization Hierarchy (from most to least preferred):

1. Source Reduction: Preventing waste generation at the source (e.g., process optimization, catalyst use,

solvent recycling).

2. Reuse/Recycling: Recovering valuable components from waste streams.

3. Treatment: Chemical, physical, or biological treatment to reduce toxicity or volume.

4. Disposal: Landfilling or incineration as a last resort.

Solvent Management: Solvent recovery and recycling are crucial due to cost and environmental impact.

7.3.3 Green Chemistry Principles and Sustainable Manufacturing Practices

Green Chemistry: A philosophy that encourages the design of products and processes that minimize the use

and generation of hazardous substances. (Developed by P. Anastas and J. Warner).

12 Principles: Prevention, atom economy, less hazardous chemical syntheses, designing safer chemicals,

safer solvents and auxiliaries, design for energy efficiency, use of renewable feedstocks, reduce derivatives,

catalysis, design for degradation, real-time analysis for pollution prevention, inherently safer chemistry for

accident prevention.

Sustainable Manufacturing: Operating processes in an economically sound manner while minimizing

negative environmental impact and conserving energy and natural resources.

Examples: Using renewable raw materials, optimizing energy use, process intensification (smaller, more

efficient reactors), closed-loop systems for water and solvent.

Relevant Environmental Regulations: Compliance with national and international environmental laws (e.g.,

Nigeria's environmental regulations, international conventions on hazardous waste).

7.4 Economics and Future Trends

7.4.1 Market Analysis, Pricing, and Supply Chain Management

Market Analysis: Understanding market size, growth drivers, competitive landscape, and customer needs.

Specialty and fine chemical markets are often niche and highly specialized.

Pricing Strategies: Value-based pricing (for specialty chemicals based on performance), cost-plus pricing,

competitive pricing. Raw material costs significantly impact profitability.

Supply Chain Management: Managing the flow of raw materials, intermediates, and finished products. For

fine/specialty chemicals, emphasis on:

Reliability: Consistent supply of high-purity raw materials.

Traceability: Especially for APIs and food additives.

Risk Management: Mitigating supply disruptions, geopolitical risks.

Sustainability: Sourcing from ethical and environmentally responsible suppliers.

7.4.2 Innovation and Emerging Technologies

Process Intensification: Developing smaller, more efficient, and safer reactors and processes (e.g., microreactors, flow chemistry).

Advanced Catalysis: Development of new, highly selective, and robust catalysts (e.g., MOFs, single-atom

catalysts, advanced biocatalysts).

Digitalization and Industry 4.0: Integration of sensors, data analytics, AI, and machine learning for process

optimization, predictive maintenance, and autonomous operation.

High-Throughput Screening: Rapidly testing many compounds or reaction conditions to accelerate discovery and optimization.

Machine Learning in Drug Discovery/Materials Design: Using AI to predict molecular properties or design new materials.

7.4.3 Role of Biotechnology and Bio-based Raw Materials

Biotechnology in Chemical Production:

Fermentation: Using microorganisms to produce chemicals (e.g., enzymes, organic acids, alcohols, some amino acids).

Biocatalysis: Using isolated enzymes for specific chemical transformations (as discussed in Module 3).

Synthetic Biology: Engineering organisms to produce novel chemicals or perform complex syntheses.

Bio-based Raw Materials (Renewable Feedstocks): Utilizing biomass (e.g., sugars, starches, lignocellulose,

vegetable oils) instead of fossil fuels as starting materials.

Benefits: Reduces reliance on finite resources, potentially lower carbon footprint.

Challenges: Cost-competitiveness, land use, scalability, processing complexity.

Examples: Bio-succinic acid, bio-based polymers (e.g., PLA from corn starch), chemicals from lignin.